Mihai Anitescu

Computational mathematics, optimization, and uncertainty quantification

e-mail: anitescu@mcs.anl.gov
home page: http://www.mcs.anl.gov/~anitescu

HONORS AND AWARDS

- Winner, 2023 INFORMS Computing Society COIN-OR Cup, for the best open-source software for optimization, for ExaModels.jl and MadNLP.jl (with Sungho Shin and Francois Pacaud).
- Winner, INFORMS Best Paper in Energy Award, 2023, for A. Subramanyam, J. Kim, M. Anitescu and others, "Failure probability constrained AC optimal power flow", IEEE Transactions on Power Systems, 37 (2022), pp. 4683-4695.
- Impact Argonne Award, 2021.
- Argonne Physical Sciences and Engineering (PSE) Excellence Award, 2020. With Charlotte Haley, Ray Osborn, and Stephan Rosenkranz
- SIAM Fellow, 2019.
- Winner, IEEE PES Best Conference Paper Award 2019 (4 winners out of 1300 submissions, with Adrian Maldonado and Michel Schanen)
- Featured, DOE's Quadrennial Technology Review, 2015. ("Improving the Energy Grid", Page 352).
- Winner, Best 2013 COAP paper award, (with Miles Lubin, Julian Hall, and Cosmin Petra)
- Finalist, Best Paper Award for the IEEE PES Summer Meeting 2014 (with S. Abhyankar and V. Rao)
- Winner of the 2013 COIN-OR cup (with Miles Lubin, Julian Hall, and Cosmin Petra)
- Recipient of the 2012 DOE INCITE Award (with a 14M CPU hour allocation on the ANL BG/L "Intrepid" Supercomputer)
- Recipient of the 2011 DOE INCITE Award (with a 10M CPU hour allocation on the ANL BG/L "Intrepid" Supercomputer)
- Wilkinson Fellow, Argonne National Laboratory, 1997-1999.
- Silver medal at the International Mathematical Olympiad, 1986.
- Prizes at the regional and national phases of The Physics Competition (both the general and the electromagnetics sections) for Romanian college students, 1989.
- Romanian National Fellowship, 1989.
- First prize at the Romanian National Mathematical Competition "Traian Lalescu" (college students), 1988.
- Multiple prizes at the regional and national phases of the Romanian National Olympiad and various other national high school mathematics and physics competitions, 1984–1986.

FUNDING

- **At Argonne External**
  - co-Investigator, "Foundation models for structured composite material discovery", DOE-NNSA, $4M total (lead Mark Messner, MSD), 2025-2028. Anitescu led the AI side at inception and transitioned to Vishwas Rao.
  - co-Investigator and thrust lead, "Energy Efficient Computing: A Holistic Methodology", DOE-ASCR Base, $15M, from FY25.
  - Lead PI "Optimal Design of Carbon Capture Units under Uncertainties in Feed Compositions and Operational Conditions", DOE-FECM through HPC4EI, $400K plus $100K cost share from Shell, 2025-2026 (option for a second phase ongoing).

- Lead PI "Macroreliability of the U.S. Power Grid: Operational Reliability of Low-Inertia Systems", DOE-OE Advanced Grid Modeling, $1,300,000, from FY24.
- Site PI "Structural Signatures of Hidden Order in Spin-Orbit Coupled Systems", DOE-BES, $750K, from FY24.
- PI "AGM Transmission Planning", DOE-OE Advanced Grid Modeling, $1.1M over three years, from FY23.
- Lead PI "Grid Modernization CASCDE", DOE Grid Modernization Initiative, $3.6M over three years, from FY23.
- Lead, one-year off-ramp extension of MACSER (MMICCS), DOE-ASCR, $500K, FY23.
- Lead PI for Argonne, "Machine-Learning Accelerated Simulations for Forecasting", DOE-ASCR, $3.2M total with $2M to Argonne, FY22-FY24.
- Lead PI "Graph-Indexed Optimization (Base)", DOE-ASCR, $1.67M, FY22-FY24.
- co-PI "Probabilistic Machine Learning for Rapid Large-scale and High-rate Aerostructure Manufacturing (PRISM)", DOE-EERE Applied Manufacturing, $2.4M, FY22-FY23 (lead Prasanna Balaprakash).
- Supplement to the Exascale ExaSGD project, $200K for Argonne, FY22-FY23.
- One-year extension, "MACRO resilience simulation to power grid", DOE-OE, $400K.
- PI "Macro-Res of the North Am Pow Grid", DOE-OE,$375K,FY20-FY22.
- PI " Accurate and Tractable Cascade Failure Models", DOE-OE,$390K,FY20-FY21.
- co-PI "Structural Signatures of Hidden Order in Spin-Orbit Coupled Systems",DOE-BES,$1M, FY20-FY22
- PI "Space-time Indexed Optimization Problems", DOE-ASCR,$440K, FY19-FY21.
- PI, "Resilience-Constrained Grid Planning", DOE-OE-AGM, $400K, 2017-2019.
- co-PI, "Bilevel Optimization Approaches for N-k Reliability Computations", $300K, 2017-2019. (PI, Kibaek Kim),
- co-PI, "Stochastic Optimization for Distribution System Uncertainty", $375K, 2017-2019 (PI, Shrirang Abhyankar),
- PI and Director, "MACSER: Multifaceted Mathematics for Complex Energy and Environmental Systems", DOE-ASCR, $ 3M, 2018-2023. At award time, the project had 27 PIs from 6 sites: Argonne, PNNL, LLNL, Univ. of Wisconsin, Univ. of Chicago, Ohio State.
- co-PI and site lead, "ECP: Stochastic Optimization at the Exascale for Planning and Cybersecurity", $ 2.5M(+1.5M,2019-2022), 2016-2022.
- PI, "Data-Driven Optimization for Complex Dynamical Systems", 440K, 2016-2019.
- PI, Grid Modernization Laboratory Consortium 1.4.18 ("Computational Science for the Grid"), 1M, 2016-2019.
- PI, "Spatio-Temporal Statistical Analysis at Scales Using Gaussian Processes", DOE-ASCR, $ 400K, 2013-2016.
- PI and Director, "M2ACS: Multifaceted Mathematics for Complex Energy Systems", DOE-ASCR, $ 3.5M, 2012-2017. At award time, the project had 22 PIs from 5 sites: Argonne, PNNL, Sandia, Univ. of Wisconsin, Univ. of Chicago.
- PI, "Scalable Dynamic Optimization", DOE-ASCR, $400K, 2012-2015.
- co-PI " Frameworks, Algorithms, and Scalable Technologies for Mathematics (FASTMath) SciDAC Institute", DOE-ASCR, $ 6.75M, 2011-2016. PI: Lori Diachin. ANL PI: Barry Smith. I am the lead PI for Differential Variational Inequalities subtopic, at $250K.
- co-PI "Climate Science for a Sustainable Energy Future (CSSEF)", DOE-BER, $ 5M 2011-2016; Lead ANL UQ PI. Lead PI: David Bader (ORNL).
- co-PI "Center for Exascale Simulation of Advanced Reactors (CESAR)", DOE-ASCR,$4M 2011-2016. Lead of the Uncertainty Quantification Area. Lead PI: Robert Rosner (University of Chicago).
- PI, "Scalable statistical analysis of Gaussian models for Petascale spatiotemporal data", DOE-ASCR,

$450K, 2009-2013.
  - PI, "Stochastic Optimization of Complex Systems", DOE-ASCR, $485K, 2009-2012.
  - PI, "Advanced Numerical Methods for Differential Variational Inequalities", $325K, DOE-ASCR, 2009-2012.
  - co-PI "SCIDAC-E: Differential Variational Inequalities for Phase Field Problems", DOE-ASCR, $300K, 2009-2011. Lead PI: Lois Curfman McInnes.
  - Co-PI, "International Symposium on Mathematical Programming", $20K, DOE-ASCR, 2009.
  - PI, " A New Challenge for Computational Science: Complementarity Constraints", $375K, DOE-ASCR, 2006-2009.
  - Co-PI, NSF DMS-0937025, "International Symposium on Mathematical Programming", $20K, 2009.
- **At Argonne External on Annual Renewal Schedules**
  - Annual Operation Plan for Officed of Energy Delivery co-PI, "Development of methodologies for large-scale optimization for electrical transmission planning". PI: Victor Zavala, 2015-2016,$300K (and 2010-2011, $375K, 2011-2012, $375K, 2012-2013, $375K, 2013-2014, $375K, 2014-2015, $375K).
  - Workpackage Manager " VV UQ for Reactor IPSC", DOE-NE, $300K, 2011–2012 (and 2010-2011, $200K).
- **At Argonne Internal**
  - Co-Investigator of the LDRD project 2025-0487, "Energy-Efficient Foundation Model for Power Grid Resilience Analysis", $50K, 2025 (PI Feng Qiu).
  - Co-Investigator of the "Big Data Grand Challenge" LDRD Award, $1.5M, 2012-2015.
  - Co-Investigator of the "Mesoscale Electro-Mechanical Materials" LDRD Award, $400K, 2012-2015.
  - Co-Investigator of the "Toward Understanding Cloud Processes and Uncertainty Modeling in Next-Generation High-Resolution Climate Models", LDRD Award, $250K, 2011-2014
  - Co-Investigator of the "Novel Power System Operations Methods for Wind-Powered Systemr" LDRD Award, by Argonne National Laboratory, $350K, 2009-2012
  - Co-Investigator of the "Separations" LDRD Award, by Argonne National Laboratory, $300K, 2007-2010
  - Co-Investigator of the "Virtual Fab Lab" LDRD Award, by Argonne National Laboratory $300K+, 2004-2006 (PIs P. Zapol and S. Gray).
- **At the University of Chicago**
  - FACCTS (France and Chicago Collaborating in the Sciences) Award, "Quantum Control", University of Chicago, $10K, 2025.
  - PI of the proposal "CPS:Synergy:GOALI:Collaborative Research: Real-time Data Analytics for Energy Cyber-Physical Systems". 384K, 2015-2018.
  - co-PI of the proposal "RTG: Computational and Applied Mathematics in Statistical Science". 1.7M, 2016-2021, PI John Lafferty.
- **At Pitt**
  - Co Principal Investigator for the National Science Foundation Grant DMS-0112239 "Scientific Computing Research Environments for the Mathematical Sciences (SCREMS)", 2001-2003, $25,000 + $39,000 University of Pittsburgh cost sharing (with J. Chadam, C. Chow, W. Layton, and I. Yotov).
  - Principal Investigator for the National Science Foundation Grant DMS-9973071, "A Computational Framework for Multi-Rigid Body Dynamics with Contact and Friction", 1999-2002, $75,000.
  - Principal Investigator for the University of Pittsburgh Grant "A Computational Framework for Multi-Rigid Body Dynamics with Contact and Friction", 1999-2001, $14,204.

TEACHING ACTIVITIES

- **University of Pittsburgh**
  - Classes taught: Elementary functions, Calculus for Engineering, Trigonometric Functions, Calcu-

lus for Business, Introduction to Linear Programming, Introduction to Numerical Linear Algebra, Introduction to Scientific Computing (graduate), Introduction to Numerical Ordinary Differential Equations (graduate), Industrial Mathematics, Numerical Linear Algebra (graduate), Introduction to Numerical Partial Differential Equations (graduate).

  - *Curriculum Development*
    * Developed the graduate class Introduction to Scientific Computing (1999).
    * Developed, together with John Burkardt of the Pittsburgh Supercomputing Center, the computational laboratory for Introduction to Scientific Computing (1999).
    * Participated in various curriculum enhancement activities of the undergraduate mathematics program, including a Sloan Foundation-sponsored development of an industrial mathematics program (2000).

- **University of Chicago, Graduate Classes**
  - Stats 31015: Computational Mathematics II: Convex Optimization, 2017, 2019
  - Stats 31020: Computational Mathematics II: Nonlinear Programming. 2014, 2016,2018,2020
  - Stats 343: Linear Models. 2012,2013,2014.
  - Stat 365: Sequential Estimation, 2016.
  - Stats 310: Computational Mathematics II: Simulation and Optimization. Winter, 2012, 2011, 2010
- **University of Chicago, Undergraduate Classes**
  - Stats 245: Statistical Theory and Methods II, 2020.
  - Math 211: Intro to Numerical Analysis

MENTORSHIP AND SUPERVISION ACTIVITIES

- **Postdoctoral scholars and long-term visitors, Argonne: 34 (3 current)**
  - April Tran (2026-present).
  - Philip Dinenis (2024-present). Research Topic: "Variational data assimilation and downscaling with data-driven surrogates" (with Vishwas Rao).
  - Mohammad S. Ramadan (2023-present). Research Topic: "Data-driven and extremum-seeking control for HPC workflows and power systems".
  - Alexis Montoison (2024-2026). Following assignment: staff at AMD.
  - Mitchell Krock (2023-2024). Subsequent position: faculty at the University of Missouri
  - Sungho Shin (2021-2024). Research Topic:" Graph-indexed Optimization". Following assignment: faculty at MIT
  - Francois Pacaud (2020-2022). Research Topic: " Condensed Methods for GPU". Following assignments: faculty, Ecole des Mines, Paris, France
  - Amanda Lenzi (2019-2022). Research Topic: "Space time statistics". Following assignment: lecturer, University of Edinburgh.
  - Anirudh Subramanyam (2018-2022). Research Topic: "Optimization under Uncertainty". Following assignment: tenure-track assistant professor at Penn State OR
  - Youngdae Kim (2018-2022). Research Topic: "Real-time Optimization". Following assignment: Scientist at Exxon-Mobil
  - Johannes Brust (2018-2021). Research Topic: "Structured quasi-Newton methods". Following assignment: lecturer at UCSD.
  - Brian Dandurand (2017-2020). Research Topic: "Stochastic Programming" (with Kibaek Kim). Following assignment: lecturer at UW-Eau Claire
  - Adrian Maldonado (2017-2019). Research Topic: "Uncertainty Propagation through Dynamic Transients." Following assignment: Staff scientist at Argonne
  - Ahmed Attia (2017-2019). Research Topic: "Data Assimilation". (with Emil Constantinescu). Following assignment: staff scientist at Argonne

- Vishwas Hebbur Vankata Rao (2017-2019). Research Topic: "Data Assimilation" (with Emil Constantinescu). Currently, staff scientist at Argonne
- Feng Qiang (2015-2017). Research Topic:"Stochastic Programming". Currently employment in industry (China)
- Michel Schanen (2015-2017). Research Topic:"Parallel Methods for Adjoint Computations". Currently, staff Scientist at Argonne
- Charlotte Haley (2014-2017). Research Topic:"Spatio-Temporal Statistics". Currently, staff Scientist at Argonne
- Francois Gilbert (2014-2017). Research Topic:"Scalable Dynamic Optimization". First employment: technical staff at Argonne. Currently: Amazon
- Julie Bessac (2014-2017). Research Topic:"Spatio-Temporal Statistics". (with Emil Constantinescu). Currently, staff Scientist at Argonne
- Hong Zhang (2014-2016). Research Topic: "Adjoints for Energy Systems". Currently, staff Scientist at Argonne
- Linyun Liang (2013-2016). Research Topic. Research Topic: "Phase Field Models". Currently Faculty at Beihang University, Beijing, China
- Zhen Zhang (2014-2015). Research Topic:"Uncertainty Quantification". Currently, Assistant Professor at UC Santa Barbara
- Ilias Bilionis (2013-2014). Research Topic. Research Topic: "Spatio-temporal statistical models for Solar and Wind Power". Currently Assistant Professor at Purdue University
- Jie Chen (2010-2014). Research Topic: "Numerical Linear Algebra for Gaussian Process Simulation". Currently, Staff Scientist at IBM
- Drew Kouri (2012-2013, Wilkinson Fellow). Research Topic: "Uncertainty Quantification and Optimization". Currently, Staff Scientist at Sandia National Laboratory.
- Jungho Lee (2010-2013, co-advised with Barry Smith). Research Topic: "Variational Inequalities in the Simulation of Heterogeneous Materials". Currently, at Citibank's Fixed Income group
- Lei Wang (2010-2013). Research Topic: "Variational Inequalities in the Simulation of Heterogeneous Materials". Currently, Assistant Professor at University of Wisconsin, Milwaukee
- Cosmin Petra (2009-2012). Research Topic: "Stochastic Programming". Currently, Computational Mathematician, Lawrence Livermore National Laboratory
- Oleg Roderick (2009-2012). Research Topic: "Hybrid Sampling-Sensitivity Approaches for Nuclear Reactors". First employment: scientist at Argonne. Currently, Data Scientist at Verizon
- Xiaoyan Zeng (2008-2012). Research Topic: "Uncertainty Analysis in Chemical Plant Safety and Climate". Currently, Assistant Professor at Shanghai University
- Emil Constantinescu (2008-2010). Research Topic: "Uncertainty Quantification". Currently, Computational Mathematician, Argonne National Laboratory
- Victor Zavala (2008-2010) Research Topic: "Optimization of Hybrid Energy Systems". Currently, Associate Professor, University of Wisconsin
- Dan Negrut (2004-2005), currently Professor at University of Wisconsin, Department of Mechanical Engineering. Research Topic: " Scalable Multiscale Methods for Orbital-Free Density Functional Theory "

- **Long-term visiting scholars, Argonne: 1**
  - Alexis Montoison (2023), student visitor
- **Full-time Argonne employees supervised: 15 (2 current)**
  - Adrian Maldonado, Energy Systems Scientist, 2019-
  - Michel Schanen, Computer Science Specialist, 2017-
  - Charlotte Haley, Statistician, 2017-2026.
  - Vishwas Rao, Computational Mathematician, 2019-2024.
  - Kibaek Kim, Assistant Computational Mathematician, 2016-2022

  - Paul Beckman, Assistant Computational Mathematician, 2019-2020 (first subsequent position, Ph.D student at NYU)
  - Francois Gilbert, Mathematics Specialist, 2017-2018. OR Researcher, Amazon
  - Julie Bessac, Statistician, 2017-2018. First subsequent position, Statistician, Argonne
  - Shrirang Abhyankar, Energy Systems Scientist, 2014-2016. Currently at ERCOT.
  - Cosmin Petra, Assistant Computational Mathematician, 2012-2016. First subsequent position: Computational Mathematician, LLNL
  - Emil Constantinescu, Assistant Computational Mathematician, Argonne National Laboratory, 2010-2015. First subsequent position: Computational Mathematician, Argonne
  - Victor Zavala, Assistant Computational Mathematician, Argonne National Laboratory, 2010-2015. First subsequent position: Computational Mathematician, Argonne
  - Jie Chen, Assistant Computational Mathematician, 2014-2014. First subsequent position: IBM
  - Oleg Roderick, Assistant Computational Mathematician, 2012-2014. Currently, Data Scientist at Geisinger Health Systems
  - Miles Lubin, Predoctoral Appointee, Argonne National Laboratory, 2011-2012 (first subsequent position: Ph.D Student at MIT)
- **Predoctoral appointees, Argonne: 6**
  - Miao Li, Predoctoral Appointee, 2022-2023 (jointly with Kibaek Kim)
  - Yuxuan Ren, Predoctoral Appointee, 2021-2022 (jointly with Kibaek Kim)
  - Albert Lam, Predoctoral Appointee, 2020-2022 (first subsequent position: Moderna)
  - Chris Geoga, Student Appointee, 2019-2022 (first subsequent position: Ph.D student at Rutgers)
  - Jisung Hwang, Predoctoral Appointee, 2020-2021 (jointly with Kibaek Kim)
  - Jacob Roth, Predoctoral Appointee, 2018-2020 (first subsequent psoition currently Ph.D at U of Minn)
  - Chris Geoga, Predoctoral Appointee, 2016-2019
- **Student interns, Argonne: 37**
  - Lin Fang (Chicago), 2026: "Pricing reliability insurance for generator portfolios under temperature-driven outage risk".
  - Lucas Jian-an Wu (Chicago, Metcalf), 2026: "Markov-chain models of generator outage and unavailable capacity versus temperature".
  - Sen Na (Chicago), 2020: "Online Nonconvex Optimization"
  - Yian Chen (Chicago), 2020: "HODLR approaches to statistical modeling"
  - Jess Kunke (Chicago), 2019: "space time statistical models for solar power"
  - Sen Na (Chicago), 2019: "Temporal Decomposition of Nonconvex Dynamic Programming"
  - Yian Chen (Chicago), 2019: "Implicit Gaussian Processes"
  - Rui Liu (Georgia Tech), 2018: "Network Optimization under Uncertainty"
  - Sen Na (Chicago), 2018: "Sensitivity of Dynamic Optimization"
  - Sungho Shin (Wisconsin) 2018: "Decentralized Control"
  - Youngseok Kim (Chicago), 2018: "SQP methods for nonparametric Bayesian estimation"
  - Richard Chen (Chicago), 2017: "Structured Linear Algebra for Gaussian Process Calculations"
  - Vivak Patel (Chicago), 2017: "Inversion of Solar Power Deployment Factors"
  - Youngseok Kim (Chicago), 2017: "SQP methods for nonparametric Bayesian estimation"
  - Shan Lu (Chicago). 2016: "Demand Models for Planning Problems"
  - Vivak Patel (Chicago). 2016: "Inversion of Dynamic Parameters of Grid"
  - Jing Yu (Chicago). 2015, 2016 "Design of Experiments for Gas Pipelines"
  - Ahmed Attia (Virginia Tech). 2014 Project "Optimization methods for importance sampling"
  - Wanting Xu (Chicago). 2014, 2015 Project " Low Memory Methods for weak 4DVar
  - Carmeline D'Silva (Princeton), 2012. Project "Uncertainty Analysis of Molecular Dynamics Simula-

tions"

- Hayes Stripling (Texas A&M University), 2011. Project " Uncertainty Analysis of Wave Reactors". Co-author of 1 journal paper
- Zhu Wang (Virginia Tech, with Oleg Roderick), 2011. Project " Using POD in Nuclear Reactor Applications"
- Miles Lubin (University of Chicago, with Cosmin Petra), 2010-2011. Project "Parallel Software for Stochastic Programming". Co-author of 1 journal paper and 1 conference proceedings paper
- Brian Lockwood (University of Wyoming, CSGF fellow), 2010. Project: " Gaussian Process Models with Derivative Information". Co-author of 1 journal paper
- Mihai Alexe, 2010. Summer student at Argonne.
- Toby Heyn (University of Wisconsin, with Dan Negrut), 2010. Project "Iterative Solvers for Differential Variational Inequalities". Co-author of 1 proceedings paper
- Zhu Wang (Virginia Tech, with Oleg Roderick), 2010. Project " Reduced Order Modeling in Nuclear Reactor Applications". Co-author of 1 journal paper
- Hammad Mazhar, 2009. Summer student at Argonne.
- Matthew Rocklin, (University of Chicago, with Emil Constantinescu), 2009. Project: "Uncertainty Quantification of Numerical Weather Prediction Systems". Co-author of 1 journal paper
- Sangmin Lee (NYU, with Victor Zavala), 2009. Project: "Stochastic Unit Commitment under Uncertainty". Co-author of 1 journal paper
- Oleg Roderick (Portland State University), 2008-2009. Project: " A Hybrid Sampling Sensitivity Approach for Uncertainty Quantification in Nuclear Reactors " co-author of 2 journal papers and 1 conference proceedings paper
- Justin Madsen, 2008. Summer student at Argonne.
- Kyle Schmitt, (MIT), 2008. Project: " Efficient Sampling of Dynamical Systems with Spatial Uncertainty " co-author of 1 conference proceedings paper and 1 journal paper
- Emil Constantinescu (Virginia Tech), 2006. Project: " Scalable Methods for Orbital-Free Density Functional Theory Calculations" Co-author of 1 conference proceedings papers
- Monika Neda (University of Pittsburgh), 2006. Project: "A collocation approach for uncertainty quantification in nuclear reactors". Co-author of 1 conference proceedings paper (ANS M&S 2007)
- Xiaoyan Zeng (Illinois Institute of Technology), 2006. Project: "Chemical Plant Safety Assessment Under Uncertainty" Co-author of 1 journal paper
- Adrian Kopacz (Northwestern), 2005. Project: " Scalable Methods for Orbital-Free Density Functional Theory Calculations"
- Bogdan Gavrea (University of Maryland, Baltimore County), 2005. Project: "Quadratic Programming Approaches for Multi-Body Dynamics with Contact and Friction". Co-author on 2 journal submissions
- Gun Srijutongsiri (grad, Cornell), 2005. Project: "Statistics of Granular Flow"
- Jufeng Peng (Rensselaer) 2003. Project: "Mathematical Programs with Complementarity Constraints in Robotics". Co-author of 1 conference proceedings paper (ICRA 2003)
- Adrian Dunca (University of Pittsburgh, with Traian Iliescu), 2002. Project: " Optimal Design of Fluid Flow Using Subproblems Reduced by Large Eddy Simulation "
- Alec Hanson (Princeton, with Lois C. McInnes), 1998. Project: "Parallel Implementation of Time Dependent Differential Variational Inequalities"
- Arman Pazouki. Summer student at Argonne.

- **Ph.D. students advised, University of Chicago: 9 (2 current)**
  - Elias Boshara, UChicago (Ph.D in progress since 2024, advised since 2025).
  - Kingsley Yeon, UChicago (Ph.D in progress since 2023, advised since 2025; research topic: structured low-rank matrix approximation).
  - Zhan Lin, UChicago (Ph.D 2024)

  - Yian Chen, Uchicago (PhD 2023)
  - Xialiang Dou, UChicago (PhD 2022, first employment: Two Sigma)
  - Sen Na, Uchicago (PhD 2021, first employment, UC Berkeley)
  - Jing Yu, UChicago, (Ph.D 2019, first employment: Citibank)
  - Vivak Patel, UChicago (Ph.D 2018, first employment: tenure-track Assistant Professor at the University of Wisconsin, Madison)
  - Wanting Xu, UChicago (with M. Stein; Ph.D 2017; first employment Morgan Stanley)
- **M.S. students advised, University of Chicago: 12**
  - Aditi Gupta, 2026, MS.
  - Guanyu Lyu, 2025, MS.
  - Yuxuan Ren, 2021, MS
  - Albert Lam, 2020, MS
  - Jess Kunke, 2020, MS
  - Jisung Hwang, 2020, MS
  - Jacob Roth, 2018, MS
  - Shan Lu, 2017, MS
  - Tom Hen, 2017, MS
  - Yuchen Su, 2017, MS
  - David Francis McDiarmid 2013, MS
  - Miles Lubin 2011, MS
- **Undergraduate theses advised, University of Chicago: 3**
  - Claire Wong, 2026.
  - Paul Beckman, 2019
  - Katie Zimmerman, 2017
- **Ph.D. students advised, University of Pittsburgh: 2**
  - Gary D. Hart, Ph.D., 2007. (with William J. Layton). Currently Instructor, University of Pittsburgh at Greensburgh. Ph.D: "A Constrained-Stabilized Time-Stepping Approach for Piecewise Smooth Multibody Dynamics. "
  - Faranak Pahlevani, Ph.D., 2004 (with William J. Layton). Ph.D Thesis: " Sensitivity Analysis of Eddy Viscosity Models". Currently Assistant Professor at Penn State Abington
- **Degree and examination committees: 3**
  - Ph.D Committees Anasthasos Liakos, Niyazi Sahin, Adrian Dunca, Hussein-Al Attas, Kimberley Jordan (all at the University of Pittsburgh mathematics department, 1999-2002), Pierre Dognin (electrical engineering, University of Pittsburgh, 2003), Arvind Pupil Raghunathan (chemical engineering, Carnegie Mellon University, 2004), Bogdan Gavrea (mathematics, University of Maryland, Baltimore County, 2006) Cosmin Petra (mathematics, University of Maryland, Baltimore County, 2009), Brian Lockwood (Wyoming, Mechanical Engineering, 2011), Joe Guinness, (U Chicago, Stats, 2012), Toby Heyn (Wisconsin, Mechanical Engineering, 2013); Hayes Stripling (Texas A &M, Nuclear Engineering, 2013); Jouke deBaar (Delft, 2014); Somak Dutta, U Chicago Stats, 2015; Michael Horell, U Chicago Stats, 2015; Jonathan Eskreis-Winkler, 2019; Yongrui Chen, 2020; Youngseok Kim, 2020;Haoyang Liu, 2020
  - MS Committess Mark Fenner, Department of Computer Science, University of Pittsburgh (2002)
  - Comprehensive Exam Committee (at Pitt 1999-2002) Students: Atife Caglar, Jon Drover, Adrian Dunca, Noel Heitmann, Faranak Pahlevani, and Niyazi Sahin, all Math Department at Pitt

REFEREEING AND EDITORIAL SERVICE

- **Editorial Boards**
  - Associate Editor, Journal of Optimization Theory and Applications (since 2025).

- Senior Editor, Optimization Methods and Software (since 2007; Software Editor 2004-2007).
- Associate Editor, SIAM Journal on Optimization (2010-2024).
- Associate Editor, SIAM/ASA Journal on Uncertainty Quantification (2012-2019).
- Associate Editor, Mathematical Programming, series A (2004-2017).
- Associate Editor, Mathematical Programming, series B (2007-2017).
- Associate Editor, SIAM Journal on Scientific Computing (2011-2016).

- **National Academy Report Review**
  - Analytic Research Foundations for the Next-Generation Electric Grid, 2016.
- **Proposal Reviews**
  - Simons Foundation Pivot Fellowship (2025).
  - The Department of Energy Early Career program (2021, 2022, 2023).
  - Kaust Research Fund (2018)
  - National Science Foundation. (1998, 1999, 2002, 2006, 2009),
  - Romanian Science Foundation (2005).
  - The Natural Science and Engineering Research Council of Canada (2004),
  - The Hong Kong Science Foundation (2002).
  - The Department of Energy (2008,2009,2010,2011,2012, 2013,2014, 2015, 2016, 2017,2018,2019).
- **Panel participation**
  - NSF Optimization panel (2024).
  - NSF Focused Research Groups (FRG) panel (2024).
  - NSF-ITR small grants panel (2002),
  - NSF Operations Research panel (2006),
  - NSF CSUMS panel (2006).
  - DOE Early Career Awards Panel (2009).
  - NSF Operation Research Panel (2015).
  - NSF Tripods Panel (2017).
  - NSF Computational Math Early Career Panel (2018)
- **Invited NSF workshop participation**
  - "Benchmarks for High Performance Computing", (2005).
  - "Mathematics in Robotics", (2000).
- **Other**
  - Refereed papers (about 15-20 a year) for SIAM Journal in Optimization, SIAM Journal Of Control, SIAM Journal Of Numerical Analysis, SIAM Journal On Scientific Computing, Numerische Mathematik, Mathematical Programming, Applied Mathematics Letters, Linear Algebra and Its Applications, Computational Optimization and Applications, Optimization Methods and Software, Control, Optimization, and the Calculus of Variations, Optimization and Engineering, Journal of Optimization Theory and Applications, ACM Transactions on Graphics, ACM SIGGRAPH, International Journal of Numerical Methods in Engineering, IEEE Transactions in Signal Processing, IEEE Transactions in Robotics, IEEE International Conference in Robotics and Automation, Transactions of the Institute of Industrial Engineers, Computers and Chemical Engineering.

PROFESSIONAL SOCIETY SERVICE

- **Professional Society Officer**
  - SAMSI Director Search Committee, 2016-2017.
  - SIAM representative to the Statistical and Applied Mathematical Sciences Institute (SAMSI) Governing Board. 1/1/15-12/31/20
  - Vice-president, the SIAM Interest group in Optimization (2011-2013).

- Vice-chair for linear programming and complementarity, Optimization Section, Institute for Operations Research and Management Science (INFORMS), 2005-2008. Organized the Linear Programming/ Complementarity Clusters at INFORMS annual meeting, 2006-2007.

- **Conference and Workshops Organization Committees**
  - ICERM Rare Events workshop, 2019.
  - IMSI Long Program "Uncertainty Quantification and AI for Complex Systems", March 3 - May 23, 2025 (organizing committee).
  - ICCOPT 2022, co-organizer of the applications track (with Christian Kirches).
  - IMSI Workshop "Expressing and Exploiting Structure in Modeling, Theory, and Computation with Gaussian Processes", 2022, co-organized with Daniel Sanz-Alonso and Michael Stein.
  - IMSI Workshop on Data Reduction in Genomics, August-September 2021 (hybrid), co-organized with M. Stephens, D. Nicolae and A. Gilbert.
  - IMSI Workshop on Verification, Validation and Uncertainty Quantification, May 2021 (online), co-organized with C. Graziani, F. Cattaneo and R. Rosner.
  - MACSER annual meeting, February 2021 (online; project lead).
  - ACNW (Argonne-Chicago-Northwestern-Wisconsin) Optimization and Machine Learning Workshop. June, 2017.
  - SIAM UQ Conferences, Lausanne, 2016
  - Optimization and UQ Workshop, IMA, Minneapolis, February 22-February 24, 2016.
  - Applied Mathematics PI meeting, Albuquerque, 2013.
  - Supercomputing 2013, member of the technical program committee.
  - Supercomputing 2015, member of the technical program committee.
  - MOS International Conference on Continuous Optimization, 2013, Portugal, Application Cluster co-organizer.
  - The 2nd International Workshop on High Performance Computing, Networking and Analytics for the Power Grid, held in conjunction with Supercomputing 2012, Nov 2012, Salt Lake City, UT.
  - The 1st International Workshop on High Performance Computing, Networking and Analytics for the Power Grid, held in conjunction with Supercomputing 2011, Nov 2011, Seattle.
  - The DOE "Mathematics for the Analysis, Simulation, and Optimization of Complex Systems" Workshop, Sep 2011, Crystal City.
  - Verification, Validation and Uncertainty Quantification Across Disciplines, 2011, Park City Utah (a workshop fully sponsored by DOE – participants got all travel and local expenses funded – through the Institute for Computation in Science).
  - Optimization in Energy Systems, 2010, Snowbird Utah (a workshop fully sponsored by DOE through the Institute for Computation in Science).
  - The DOE Crosscut Extreme Scale Workshop, 2010, Rockville, MD.
  - The 2nd INL/NCSU Verification and Validation Workshop, Myrtle Beach, SC, 2010.
  - ICCOPT 2010: The 3rd International Conference on Continuous Optimization of the Mathematical Programming Society, Santiago de Chile.
  - SIAM Annual Meeting, 2009, Denver
  - International Symposium for Mathematical Programming, 2009, Chicago.
  - Steering Committee, Midwest Numerical Analysis Conference, (2007-present).
  - The 15-th, 16-th, 17-th, 18-th International Conference on Control Systems and Computer Science, Bucharest, Romania, 2005, 2007, 2009, 2011 (sponsored by the regional IEEE chapter).
  - Joint EUROPT-OMS Meeting 2007: 2nd Conference on Optimization Methods and Software and 6th EUROPT Workshop on Advances in Continuous Optimization July 4-7, 2007, in Prague, Czech Republic (member of program committee).
  - World Congress in Computational Mechanics, 2006, Los Angeles (member of advisory committee)

- **Minisymposia organized**

- Characterization and Prediction of Rare and Extreme Events in Complex Systems, SIAM Conference on Computational Science and Engineering, 2021 (two-part minisymposium).
- Optimization INFORMS fall meeting, San Antonio, 2000; INFORMS fall meeting, San Jose 2002 (2); International Symposium in Mathematical Programming, Copenhagen, 2003; INFORMS fall meeting, Atlanta 2003 (2); Canadian Operations Research Society international meeting, Banff, 2004; INFORMS fall meeting, Denver 2004 (3), International Conference on Complementarity Problems, Stanford 2005; INFORMS fall meeting San Francisco 2005 (2); 2-nd International Conference on Continuous Optimization, Hamilton, Canada, 2007; INFORMS Annual Meeting 2007, Seattle; INFORMS Annual Meeting Washington DC, 2008; INFORMS Annual Meeting 2009, San Diego (2); SIAM CSE Conference (2011, 2); SIAM Parallel Processing Conference, 2012; International Symposium in Mathematical Programming, 2012 (3), Berlin; SIAM CSE Conference, Boston, 2013, SIAM UQ Conference, Savannah 2014, SIAM Optimization Conference, San Diego 2014, SIAM Annual Meeting Chicago 2014; SIAM UQ Conference; SIAM Optimization 2017, Vancouver (3); SIAM MPE Philadelphia 2018; SIAM CSE Spokane 2019; Mathematical Optimization of Systems Impacted by Rare, High-Impact Random Events, ICERM 2019, Providence RI,
- Multibody Dynamics World Congress in Computational Mechanics, Los Angeles, 2006; International Conference for Intelligent Robots and Systems, 2003 (Dynamics Section).
- Uncertainty Quantification Idaho Verification and Validation Workshop, Myrtle Beach, (2010), SIAM UQ Conference, Raleigh, 2012;
- Artificial Intelligence Energy-Fusion Energy Sciences Breakout at the ANL 2019 Artificial Intelligence Meeting (with Bill Tang); Smart Energy Infrastructure track at the DOE Town Hall Meeting in AI, Wash DC 2019 (with Teja Kuruganti, Mihai Anitescu, Mary Ann Piette, Tianzhen Hong).

PRIZE AND SPECIAL LECTURE COMMITTEE WORK

- OMS Broyden Prize Committee 2022 (Chair)
- OMS Broyden Prize Committee 2021.
- SIAM UQ Officer Nomination Committee, 2022.
- SIAM SIAG/OPT Best Paper Prize Committee, 2020 (Chair)
- Mathematical Optimization Society – Society of Industrial and Applied Mathematics (MOS-SIAM) 2015 Lagrange Prize Committee (chair).
- International Congress of Mathematicians ICM 2014, Control and Optimization Panel.
- Society for Industrial and Applied Mathematics (SIAM) Optimization Activity Group Prize Committee, 2013 (chair).
- INFORMS Coin-OR Cup Prize Committee, 2014.
- INFORMS Coin-OR Cup Prize Committee, 2025 (Chair).
- OMS Broyden Best Paper Prize Committee, 2024-2025.
- SIAM UQ Early Career Prize Committee, 2021.

COMMITTEE WORK

- **University of Chicago**
  - Computational and Applied Mathematics Graduate Admission Committee, 2018-
  - Computational and Applied Mathematics Undergraduate Major Steering Committee 2014-.
  - Computational and Applied Mathematics Initiative Hiring Committee, 2012-2014.
- **Argonne National Laboratory**
  - MCS (divisional) strategic council (2018-)
  - Co-chaired the Data Forum (2012-2014)
  - Chaired the Division Director Search Committee (2010-2011).
  - Director's Postdoctoral Committee (2010-2012).

- Chaired the Wilkinson Selection Committee (2012, 2010, 2008).
- Divisional Awards Committee (2009,2010).

- **University of Pittsburgh**
  - Served in the Mellon Chair Search Committee (1999).
  - Departmental Chair Selection Committee (2000).
  - Scientific Computing Hiring Committee (2002).

INVITED TALKS

- **2026-2030**
  - "Exponential Decay of Sensitivity in Dynamic Programming and Computational Consequences", Numerical Analysis, Approximation Theory and Stochastic Methods Conference, Cluj Napoca, Romania, September 2026 (Plenary)
  - "Scalability and Sustainability Tools for Digital Twins from Power Grid Applications", Fermi National Lab Colloquium, July 2026
  - "Optimization for Estimation of Rare Event Probabilities in Energy Systems", SIAM Optimization Conference, Edinburgh, June 2026
  - "Exponential Decay of Sensitivity in Dynamic Programming and Computational Consequences", National University of Singapore, March 2026
  - "PIONEER-OUU thrust", Applied Math PI meeting, July 2026
  - "TEMPEST: Tool for sEvere weather Mitigation, Predictive Evaluation, and System Transformation", NERC Probabilistic Working Group Online Seminar, January 2026
- **2021-2025**
  - "Optimization for Digital Twins of Weather and Power", New York Academy of Sciences panel, September 2025
  - "Scalability and Sustainability Tools for Digital Twins from Power Grid Applications", Colloquium, Flatiron Institute, September 2025
  - "Scalable Multi-Period ACOPF utilizing GPUs with High Memory Capacities", International Conference on Continuous Optimization, Los Angeles, July 2025
  - "Exponential Decay of Sensitivity in Dynamic and other Graph-Indexed Programming and Computational Consequences", USC Operations Research Colloquium, April 2025
  - "Tail dependence as a measure of teleconnected warm and cold extremes of North American wintertime temperatures", SIAM Conference on Computational Science and Engineering, 2025
  - "Reliability, Extreme Weather Events, and Downscaling for Energy Applications", Notre Dame Statistics Colloquium, March 2025
  - "Reliability, Extreme Weather Events, and Downscaling for Energy Applications", NYU Tandon School of Engineering Workshop on Decision under Uncertainty for Energy, January 2025
  - "Issues in Modeling and Computing with Extremes", NSF Workshop on Data-driven Modeling and Prediction of Rare and Extreme Events, Institute for Mathematical and Statistical Innovation, November 2024
  - "GPU-accelerated nonlinear programming", INFORMS Annual Meeting, Seattle, October 2024
  - "Condensed Space Methods for Nonlinear Programming on GPU", International Symposium on Mathematical Programming, Montreal, July 2024
  - "Large Deviation Approaches for Resilience Calculations", IMSI Green Energy Workshop, July 2024
  - "Condensed Space Methods for Nonlinear Programming on GPU", NREL 7th Autonomous Systems Workshop, 2024
  - "The Two Language Problem; Productivity in Technical Computing; Domain Specific Languages; Rapid Digital Twins", University of Chicago Statistics Seminar, December 2023
  - "Exponential Decay of Sensitivity in Graph-Indexed Optimization", IIT seminar, September 2023

- "Rapid Digital Twins in Power Grid", National Academies Symposium on Digital Twins, July 2023 (virtual)
- "Scalable Interior Point Algorithms on GPU", FERC software conference, June 2023
- "Exponential Decay of Sensitivity in Graph-Indexed Optimization", SIAM Conference on Optimization, May 2023
- "Rare Events Computation in Power Grid and Climate", Clemson Extreme Event workshop, May 2023
- "Exponential Decay of Sensitivity in Graph-Indexed Optimization", 12th US-Mexico Workshop on Optimization and its Applications, January 2023
- "Scalable Gaussian Process Calculations", RWTH Aachen, June 2022 (remote seminar)
- "Scalable Gaussian Process Calculations", ORNL AIRES Workshop, April 2022 (remote)
- "Rare event simulation of energy cascades using Kinetic Monte Carlo", IEEE PES Society-Wide seminar, October 2021
- "Exponential Decay of Sensitivity in Dynamic Programming and Computational Consequences", US Naval Academy Conference in Optimization, June 2021
- "Rare event simulation and control of electricity cascades", PHILMS virtual seminar, April 2021
- "Rare event simulation and control of electricity cascades", SIAM Conference on Computational Science and Engineering, 2021 (virtual)
- "Exponential Decay of Sensitivity in Dynamic Programming and Computational Consequences", IMSI workshop on Digital Twin Control, October 2025
- "TEMPEST: Tool for sEvere weather Mitigation, Predictive Evaluation, and System Transformation", presentation to NERC potential clients, October 2025

- **2016-2020**
  - IMA Seminar on Industrial Mathematics (2016)
  - AMS Meeting, Denver, (2016)
  - FERC Software Conference (Washington, DC, June 2016)
  - SAMSI Optimization Workshop, Research Triangle Park, August 2016
  - SIAM CSE, Atlanta (2017)
  - SIAM Optimization Conference (2017)
  - Applied Math Colloquium at U of Iowa (2017)
  - Computational Science and Engineering Seminar (Notre Dame, 2018)
  - The 9th annual graduate student mini-conference in Computational Mathematics, Columbia, SC (2018, keynote)
  - 7th International Conference on High Performance Scientific Computing, Hanoi, Vietnam (2018, plenary)
  - Purdue Data Science Day, Purdue, IN (2018)
  - SIAM UQ Conference, Orange Grove, CA (invited, 2018)
  - IMA Workshop on Forecasting for Complex Processes (invited, 2018)
  - The International Symposium on Mathematical Programming, Bordeaux, France (invited, 2018)
  - INFORMS annual meeting 2018 (invited)
  - SIAM Conference on Mathematics of Planet Earth, 2018 (invited panelist)
  - IEEE PES 2019 (invited panelist by the UW working group)
  - Computing in Engineering Forum 2019, Wisconsin Madison (invited)
  - Computing in Engineering Forum 2020, (virtual, invited)
  - IEEE PES HPC Working Group Webinar, 2020 (invited, virtual)
  - IEEE PES Webinar, 2020 (invited).
- **2011-2015**
  - SIAM Computational Science and Engineering Conference, Reno (2011)

- SIAM Optimization Meeting, Darmstadt, Germany, (2011)
- International Conference in Control Systems and Computer Science, Bucharest, Romania (plenary, 2011)
- IMA Workshop on UQ, Minneapolis, (2011)
- SAMSI Summer School on UQ, Albuquerque (invited lecturer, 2011)
- The DOE Applied Math Research PI Meeting 2011, (plenary)
- University of Texas, ICES, 2011 (Seminar)
- University of Pittsburgh 2012
- SIAM UQ Conference 2012
- University of Chicago 2012
- the International Symposium on Mathematical Programming, Berlin, 2012
- University of Illinois (the IMSE kick-off conference), 2012
- DTRA CyberPhysical Networks Workshop, Washington DC, 2012
- International Conference on Complementarity Problems, Singapore, 2012
- Powertech 2013, Grenoble, 2013
- ACCES Workshop, Aachen, 2013
- SIAM Optimization Conference, San Diego, 2014
- SIAM UQ Conference, Savannah 2014
- International Conference on Complementarity Problems, Berlin, 2014
- ACNW Workshop, Chicago 2015
- NREL, Golden, 2015
- IMSP Pittsburgh 2015
- INFORMS, Philadelphia 2015 (keynote)
- EPRI Innovation Summit, Huntington Beach, October 2015
- ABB Colloquium, Raleigh, December 2015

- **2006-2010**
  - Illinois Institute of Technology (2006)
  - Workshop on Assessment of Sensitivity/Uncertainty Analysis Capabilities Applicable for the Nuclear Fuel Cycle, North Carolina State University (2006)
  - 3-rd Workshop on the Global Nuclear Energy partnership, Washington, D.C. (2006)
  - International Congress on Industrial and Applied Mathematics, Zurich (2007)
  - International Conference on Continuous Optimization, Hamilton, (semi-plenary, 2007)
  - Computational and Mathematical Methods in Science and Engineering, (plenary, 2007)
  - International Workshop on Hybrid Systems Modeling, Simulation and Optimization Koc University, Istanbul (2008)
  - GNEP Verification and Validation workshop, Idaho Falls (2008)
  - Stevens Institute of Technology Colloquium (2008)
  - INFORMS Annual Meeting, Washington DC (2008)
  - SIAM Conference on Computational Science and Engineering, Miami (2009)
  - University of Illinois at Chicago Math Colloquium (2009)
  - American Nuclear Society Annual Meeting, Atlanta (2009, roundtable keynote)
  - International Symposium on Mathematical Programming, Chicago (2009, semi-plenary)
  - SIAM Annual Meeting, Denver, (2009)
  - US National Congress in Computational Mechanics, Columbus, (2009)
  - INFORMS Annual Meeting (2009)
  - Lehigh High Performance Computing Meeting (2009, plenary)
  - SIAM Parallel Processing Conference, Seattle (2010)

- ANS Annual Meeting San Diego (2010)
- 2nd Verification and Validation Workshop, Myrtle Beach (2010)
- 12-th International Conference on Stochastic Programming, Halifax, (2010)
- Princeton University, Operations Research and Financial Engineering Colloquium (2010)
- University of Wisconsin, Computer Science Colloquium (2010)

- **2001-2005**
  - Old Dominion University (2001)
  - IMA Workshop on Haptics, Virtual Reality and Human Computer Interaction, Minneapolis (2001)
  - Contact Mechanics International Symposium, Peniche, Portugal, (2001)
  - International Conference on Scientific Computation And Differential Equations, Vancouver, Canada (2001)
  - INFORMS fall meeting, Miami Beach (2001)
  - SIAM Optimization Conference, Toronto, (2002)
  - International Symposium on Complementarity Problems, Cambridge, (2002)
  - INFORMS fall meeting, San Jose, (2002)
  - International Congress on Computational and Applied Mathematics, Sydney (2003)
  - SIAM Annual Meeting, Montreal (2003)
  - INFORMS fall meeting, Atlanta (2003)
  - International Conference on Intelligent Robots and Systems, Las Vegas (2003)
  - SIAM Annual Meeting, Portland (2004)
  - Sandia Multiscale Optimization Workshop (2004)
  - BIRS Workshop on Molecular Dynamics, Banff (2005)
  - Midwest Numerical Analysis Conference, (plenary, 2005)
  - 15th International Conference on Control and Computer Science, Bucharest, (2005)
  - 4th International Conference on Complementarity Problems, Stanford, (2005)
- **1997-2000**
  - Georgia Institute of Technology (1997)
  - Clemson University (1997)
  - Argonne National Laboratory (1997)
  - University of Louisville (1997)
  - University of Pittsburgh (1997)
  - University of Iowa (1997)
  - SIAM Conference in Optimization, Atlanta (1999)
  - INFORMS fall meeting, Philadelphia (1999)
  - University of Maryland, Baltimore County (1999)
  - INFORMS spring meeting, Salt Lake City (2000)
  - Mathematical Programming Symposium, Atlanta (2000)
  - INFORMS fall meeting, San Antonio (2000)

PUBLICATIONS

JOURNAL PAPERS PUBLISHED OR IN PRINT

1. Pacaud, François, Sungho Shin, Alexis Montoison, Michel Schanen, and Mihai Anitescu. "Condensed interior-point methods for scalable nonlinear programming on GPUs." Mathematical Programming Computation (2026). doi:10.1007/s12532-026-00335-0.
2. Ramadan, Mohammad S., Evan Toler, and Mihai Anitescu. "Data-Conforming Data-Driven Control: Avoiding Premature Generalizations Beyond Data." IEEE Transactions on Automatic Control (2026).

doi:10.1109/tac.2025.3647312.

3. Shin, Sungho, Sen Na, and Mihai Anitescu. "Near-Optimal Performance of Stochastic Model Predictive Control." Mathematics of Operations Research (2026). doi:10.1287/moor.2023.0159.

4. Kuang, Wei, Mihai Anitescu, and Sen Na. "Online covariance matrix estimation in sketched Newton methods." Information and Inference: A Journal of the IMA (2026). doi:10.1093/imaiai/iaag012.

5. Dobson, Ian, D. Adrian Maldonado, and Mihai Anitescu. "The Statistical Spread of Transmission Outages on a Fast Protection Time Scale Based on Utility Data." IEEE Transactions on Power Systems (2026). doi:10.1109/tpwrs.2025.3596141.

6. Zhao, Hongli, Mihai Anitescu, and Sen Na. "Overlapping Schwarz Scheme for Linear-Quadratic Programs in Continuous Time." Mathematics of Operations Research (2026). doi:10.1287/moor.2025.1232.

7. Rossmann, Ramsey, Mihai Anitescu, Julie Bessac, Michael Ferris, Mitchell Krock, James Luedtke, and Line Roald. "A framework for balancing power grid efficiency and risk with bi-objective stochastic integer optimization." Optimization and Engineering (2025). doi:10.1007/s11081-024-09944-x.

8. Kim, Youngdae, François Pacaud, Michel Schanen, Kibaek Kim, and Mihai Anitescu. "Leveraging GPU Batching for Scalable Nonlinear Programming through Massive Lagrangian Decomposition." SIAM Journal on Scientific Computing (2025). doi:10.1137/21m1450112.

9. Anderson, Edward, Michael Ferris, Andrew Philpott, Mihai Anitescu, Peter Cramton, Sijia Geng, Richard Green, Tito Homem-de-Mello, Olivier Huber, Vincent Leclère, and Ramteen Sioshansi. "Ten challenges for mathematical modeling of the green-energy transition." Current Sustainable/Renewable Energy Reports (2025). doi:10.1007/s40518-025-00274-9.

10. Na, Sen, Mihai Anitescu, and Mladen Kolar. "A Fast Temporal Decomposition Procedure for Long-Horizon Nonlinear Dynamic Programming." Mathematics of Operations Research (2024). doi:10.1287/moor.2023.1378.

11. Shin, Sungho, Mihai Anitescu, and François Pacaud. "Accelerating optimal power flow with GPUs: SIMD abstraction of nonlinear programs and condensed-space interior-point methods." Electric Power Systems Research (2024). doi:10.1016/j.epsr.2024.110651.

12. Bhaduri, Anindya, Nesar Ramachandra, Sandipp Krishnan Ravi, Lele Luan, Piyush Pandita, Prasanna Balaprakash, Mihai Anitescu, Changjie Sun, and Liping Wang. "Efficient Mapping Between Void Shapes and Stress Fields Using Deep Convolutional Neural Networks With Sparse Data." Journal of Computing and Information Science in Engineering (2024). doi:10.1115/1.4064622.

13. Ramadan, Mohammad S., and Mihai Anitescu. "Extended Kalman Filter–Koopman Operator for Tractable Stochastic Optimal Control." IEEE Control Systems Letters (2024). doi:10.1109/lcsys.2024.3410889.

14. Pacaud, François, Michel Schanen, Sungho Shin, Daniel Adrian Maldonado, and Mihai Anitescu. "Parallel interior-point solver for block-structured nonlinear programs on SIMD/GPU architectures." Optimization Methods and Software (2024). doi:10.1080/10556788.2024.2329646.

15. Pacaud, François, Sungho Shin, Michel Schanen, Daniel Adrian Maldonado, and Mihai Anitescu. "Accelerating Condensed Interior-Point Methods on SIMD/GPU Architectures." Journal of Optimization Theory and Applications (2023). doi:10.1007/s10957-022-02129-5.

16. Lenzi, Amanda, and Mihai Anitescu. "How can statistics help to prevent blackouts?." Significance (2023). doi:10.1093/jrssig/qmad009.

17. Na, Sen, Mihai Anitescu, and Mladen Kolar. "Inequality constrained stochastic nonlinear optimization via active-set sequential quadratic programming." Mathematical Programming (2023). doi:10.1007/s10107-023-01935-7.

18. Shin, Sungho, Yiheng Lin, Guannan Qu, Adam Wierman, and Mihai Anitescu. "Near-Optimal Distributed Linear-Quadratic Regulator for Networked Systems." SIAM Journal on Control and Optimization (2023). doi:10.1137/22m1489836.

19. Chen, Yian, and Mihai Anitescu. "Scalable Physics-Based Maximum Likelihood Estimation Using Hierarchical Matrices." SIAM/ASA Journal on Uncertainty Quantification (2023). doi:10.1137/21m1458880.

20. Beckman, Paul G., Christopher J. Geoga, Michael L. Stein, and Mihai Anitescu. "Scalable computations for nonstationary Gaussian processes." Statistics and Computing (2023). doi:10.1007/s11222-023-10252-0.

21. Na, Sen, and Mihai Anitescu. "Superconvergence of Online Optimization for Model Predictive Control." IEEE Transactions on Automatic Control (2023). doi:10.1109/tac.2022.3223323.

22. Shin, Sungho, Mihai Anitescu, and Victor M. Zavala. "Exponential decay of sensitivity in graph-structured nonlinear programs." SIAM Journal on Optimization 32.2 (2022): 1156-1183. Awarded the 2023 W. David Smith, Jr. Graduate Student Paper Award given by the Computing and Systems Technology Division of the AIChE; to Sungho Shin.

23. Pacaud, François, Daniel Adrian Maldonado, Sungho Shin, Michel Schanen, and Mihai Anitescu. "A feasible reduced space method for real-time optimal power flow." Electric Power Systems Research (2022). doi:10.1016/j.epsr.2022.108268.

24. Na, Sen, Mihai Anitescu, and Mladen Kolar. "An adaptive stochastic sequential quadratic programming with differentiable exact augmented lagrangians." Mathematical Programming (2022). doi:10.1007/s10107-022-01846-z.

25. Brust, Johannes J., and Mihai Anitescu. "Convergence Analysis of Fixed Point Chance Constrained Optimal Power Flow Problems." IEEE Transactions on Power Systems (2022). doi:10.1109/tpwrs.2022.3146873.

26. Lenzi, Amanda, Julie Bessac, and Mihai Anitescu. "Detecting Large Frequency Excursions in the Power Grid With Bayesian Decision Theory." IEEE Open Access Journal of Power and Energy (2022). doi:10.1109/oajpe.2022.3144612.

27. Tan, Bendong, Junbo Zhao, Nan Duan, Daniel Adrian Maldonado, Yingchen Zhang, Hongming Zhang, and Mihai Anitescu. "Distributed Frequency Divider for Power System Bus Frequency Online Estimation Considering Virtual Inertia From DFIGs." IEEE Journal on Emerging and Selected Topics in Circuits and Systems (2022). doi:10.1109/jetcas.2022.3141975.

28. Subramanyam, Anirudh, Jacob Roth, Albert Lam, and Mihai Anitescu. "Failure Probability Constrained AC Optimal Power Flow." IEEE Transactions on Power Systems (2022). doi:10.1109/tpwrs.2022.3146377.

29. Lenzi, Amanda, and Mihai Anitescu. "How Statistics Keeps the Lights on." Significance (2022). doi:10.1111/1740-9713.01704.

30. Rao, Vishwas, Charlotte Haley, and Mihai Anitescu. "LaplaceInterpolation.jl: A Julia package for fast interpolation on a grid." Journal of Open Source Software (2022). doi:10.21105/joss.03766.

31. Na, Sen, Sungho Shin, Mihai Anitescu, and Victor M. Zavala. "On the Convergence of Overlapping Schwarz Decomposition for Nonlinear Optimal Control." IEEE Transactions on Automatic Control (2022). doi:10.1109/tac.2022.3194087.

32. Upreti, Puspa, Matthew Krogstad, Charlotte Haley, Mihai Anitescu, Vishwas Rao, Lekh Poudel, Omar Chmaissem, Stephan Rosenkranz, and Raymond Osborn. "Order-Disorder Transitions in (CaxSr1-x)3Rh4Sn13." Physical Review Letters (2022). doi:10.1103/physrevlett.128.095701.

33. Maldonado, Daniel Adrian, Emil Mihai Constantinescu, Hong Zhang, Vishwas Rao, and Mihai Anitescu. "Trust-Region Approximation of Extreme Trajectories in Power System Dynamics." IEEE Transactions on Power Systems (2022). doi:10.1109/tpwrs.2022.3141372.

34. Schanen, M, V Churavy, Y Kim, and M Anitescu. "Rapid Prototyping with Julia: From Mathematics to Fast Code." Collections 55 (06) (2022).

35. Rao, Vishwas, and Mihai Anitescu. "Efficient Computation of Extreme Excursion Probabilities for Dynamical Systems through Rice's Formula." SIAM/ASA Journal on Uncertainty Quantification (2021). doi:10.1137/19m1308177.

36. Lenzi, Amanda, Julie Bessac, and Mihai Anitescu. "Power grid frequency prediction using spatiotemporal modeling." Statistical Analysis and Data Mining: The ASA Data Science Journal (2021). doi:10.1002/sam.11535.

37. Chen, Yian, and Mihai Anitescu. "SCALABLE GAUSSIAN PROCESS ANALYSIS FOR IMPLICIT PHYSICS-BASED COVARIANCE MODELS." International Journal for Uncertainty Quantification (2021). doi:10.1615/int.j.uncertaintyquantification.2021036657.

38. Geoga, Christopher J., Mihai Anitescu, and Michael L. Stein. "Flexible nonstationary spatio-temporal modeling of high-frequency monitoring data". To appear in Environmetrics. arXiv preprint arXiv:2007.11418 (2020).

39. Na, Sen, and Mihai Anitescu. "Exponential decay in the sensitivity analysis of nonlinear dynamic programming." SIAM Journal on Optimization 30, no. 2 (2020): 1527-1554.

40. Shin, Sungho, Victor M. Zavala, and Mihai Anitescu. "Decentralized schemes with overlap for solving graph-structured optimization problems." IEEE Transactions on Control of Network Systems 7, no. 3 (2020): 1225-1236.

41. Geoga, Christopher J., Mihai Anitescu, and Michael L. Stein. "Scalable Gaussian process computations using hierarchical matrices." Journal of Computational and Graphical Statistics 29, no. 2 (2020): 227-237.

42. Kim, Youngseok, Peter Carbonetto, Matthew Stephens, and Mihai Anitescu. "A fast algorithm for maximum likelihood estimation of mixture proportions using sequential quadratic programming." Journal of Computational and Graphical Statistics 29, no. 2 (2020): 261-273.

43. Schanen, Michel, Daniel Adrian Maldonado, François Pacaud, Alexis Montoison, Mihai Anitescu, Kibaek Kim, Youngdae Kim, Vishwas Rao, and Anirudh Subramanyam. "Julia as a portable high-level language for numerical solvers of power flow equations on GPU architectures." 2020. doi:10.13140/rg.2.2.22427.39205.

44. Roth, Jacob, David A. Barajas-Solano, Panos Stinis, Jonathan Weare, and Mihai Anitescu. "A Kinetic Monte Carlo Approach for Simulating Cascading Transmission Line Failure." To appear in SIAM Journal on Multiscale Modeling and Simulation. arXiv preprint arXiv:1912.08081 (2019).

45. Xu, Wanting, and Mihai Anitescu. "Exponentially convergent receding horizon strategy for constrained optimal control." Vietnam Journal of Mathematics 47, no. 4 (2019): 897-929.

46. Yu, Jing, and Mihai Anitescu. "Multidimensional sum-up rounding for integer programming in optimal experimental design." Mathematical Programming (2019): 1-40.

47. Dou, Xialiang, and Mihai Anitescu. "Distributionally robust optimization with correlated data from vector autoregressive processes." Operations Research Letters 47, no. 4 (2019): 294-299.

48. Petra, Cosmin G., Naiyuan Chiang, and Mihai Anitescu. "A structured quasi-newton algorithm for optimizing with incomplete hessian information." SIAM Journal on Optimization 29, no. 2 (2019): 1048-1075.

49. Xu, Wanting, and Mihai Anitescu. "Exponentially accurate temporal decomposition for long-horizon linear-quadratic dynamic optimization." SIAM Journal on Optimization 28, no. 3 (2018): 2541-2573.

50. Geoga, Christopher J., Charlotte L. Haley, Andrew R. Siegel, and Mihai Anitescu. "Frequency–wavenumber spectral analysis of spatio-temporal flows." Journal of Fluid Mechanics 848 (2018).

51. Linyun Liang, Zhi-Gang Mei, Yeon Soo Kim, Mihai Anitescu, and Abdellatif M. Yacou Three-dimensional phase-field simulations of intragranular gas bubble evolution in irradiated U-Mo fuel. Computational Materials Science, Volume 145, 2018, Pages 86-95.

52. Julie Bessac, Emil M. Constantinescu and Mihai Anitescu. "Stochastic simulation of predictive space-time scenarios of wind speed using observations and physical models, ", "Annals of Applied Statistics", Volume 12, Number 1 (March 2018), 432-458.

53. Yu, Jing, Victor M. Zavala, and Mihai Anitescu. "A scalable design of experiments framework for optimal sensor placement." Journal of Process Control, Volume 67, Pages 44-55, July 2018.

54. C. L. Haley and M. Anitescu, "Optimal Bandwidth for Multitaper Spectrum Estimation," in IEEE Signal Processing Letters, vol. 24, no. 11, pp. 1696-1700, Nov. 2017.

55. Abhyankar, Shrirang; Geng, Guangchao; Anitescu, Mihai; Wang, Xiaoyu; Dinavahi, Venkata: 'Solution techniques for transient stability-constrained optimal power flow Part I, IET Generation, Transmission &; Distribution, 2017, 11, (12), p. 3177-3185.

56. Zhang, Hong, Shrirang Abhyankar, Emil Constantinescu, and Mihai Anitescu. "Discrete Adjoint Sensitivity Analysis of Hybrid Dynamical Systems With Switching." IEEE Transactions on Circuits and Systems I: Regular Papers 64, no. 5 (2017): 1247-1259.

57. Petra, Noemi, Cosmin G. Petra, Zheng Zhang, Emil M. Constantinescu, and Mihai Anitescu. "A Bayesian approach for parameter estimation with uncertainty for dynamic power systems." IEEE Transactions on Power Systems 32, no. 4 (2017): 2735-2743. Preprint ANL/MCS-P5525-0116.

58. Anitescu, Mihai, Jie Chen, and Michael L. Stein. "An inversion-free estimating equations approach for gaussian process models." Journal of Computational and Graphical Statistics 26.1 (2017): 98-107.

59. Zavala, Victor M., Kibaek Kim, Mihai Anitescu, and John Birge. "A stochastic electricity market clearing formulation with consistent pricing properties." Operations Research 65, no. 3 (2017): 557-576.

60. Linyun Liang, Zhi-Gang Mei,, Yeon Soo Kim, Bei Ye, Gerard Hofman, Mihai Anitescu, Abdellatif M. Yacout. "Mesoscale model for fission-induced recrystallization in U-7Mo alloy" Computational Materials Science Volume 124, November 2016, Pages 228-237.

61. Xu, Wanting, and Mihai Anitescu. "A Limited-Memory Multiple Shooting Method for Weakly Constrained Variational Data Assimilation." SIAM Journal on Numerical Analysis 54, no. 6 (2016): 3300-3331

62. Petra, Cosmin G., Victor M. Zavala, Elias D. Nino-Ruiz, and Mihai Anitescu. "A high-performance computing framework for analyzing the economic impacts of wind correlation." Electric Power Systems Research 141 (2016): 372-380.

63. Zhi-Gang Mei*, Linyun Liang, Yeon Soo Kim, Tom Wiencek, Edward O'Hare, Abdellatif M. Yacout, Gerard Hofman, Mihai Anitescu. Grain growth in U-7Mo alloy: a combined first-principles and phase field study. Journal of Nuclear Materials Volume 473, May 2016, Pages 300

64. Linyun Liang, Marius Stan and Mihai Anitescu. "Phase-field modeling of diffusion-induced crack propagations in electrochemical systems". Appl. Phys. Lett. 105, 163903 (2014).

65. Ilias Bilionis, Emil M. Constantinescu, Mihai Anitescu. "Data-Driven Model for Solar Irradiation Based on Satellite Observations". Solar Energy.Volume 110, December 2014, Pages 22–38. Preprint ANL/MCS-P5141-0514

66. Cosmin G. Petra, Olaf Schenk, Mihai Anitescu. "Real-time Stochastic Optimization of Complex Energy Systems on High Performance Computers". Computing in Science and Engineering, Issue No.05 - Sept.-Oct. (2014 vol.16) pp: 32-42. Preprint ANL/MCS-P4068-0413.

67. Victor M. Zavala and Mihai Anitescu. "Scalable Nonlinear Programming Via Exact Differentiable Penalty Functions And Trust-Region Newton Method",SIAM Journal on Optimization, 24(1), 528–558, 2014. Preprint ANL/MCS-P3014-0712.

68. Jie Chen, Tom L. H. Li and Mihai Anitescu. A Parallel Linear Solver for Multilevel Toeplitz Systems with Possibly Several Right-Hand Sides. Parallel Computing, Volume 40, Issue 8, August 2014, Pages 408-424.

69. Xiaoyan Zeng and Mihai Anitescu. "Sequential Monte Carlo Sampling in Hidden Markov Models of Nonlinear Dynamical Systems".Applied Mathematics and Computation 233 (2014) 507–521. ALso Preprint ANL/MCS-P1852-0311.

70. Oleg Roderick, Mihai Anitescu and Yulia Peet. "Proper orthogonal decompositions in multifidelity uncertainty quantification of complex simulation model". International Journal of Computer Mathematics Volume 91, Issue 4, 2014, 748-769. Preprint ANL/MCS-P4029-011.

71. Mihai Anitescu, Xiaoyan Zeng and Emil Constantinescu. "A Low-Memory Approach For Best-State Estimation Of Hidden Markov Models With Model Error". SIAM J. NUMER. ANAL. Vol. 52, No. 1, pp. 468–495, 2014. Preprint ANL/MCS-P1919-0711.

72. Jie Chen, Lei Wang, And Mihai Anitescu. "A Fast Summation Tree Code For Matern Kernel", SIAM J. SCI. COMPUT., Vol. 36, No. 1, pp. A289–A309, 2014. Also Preprint ANL/MCS-P4001-1212.

73. Toby Heyn, Mihai Anitescu, Alessandro Tasora and Dan Negrut. "Using Krylov Subspace and Spectral Methods for Solving Complementarity Problems in Many-Body Contact Dynamics Simulation", International Journal for Numerical Methods in Engineering, Volume 95, Issue 7, pages 541–561, 17 August 2013. Preprint ANL/MCS-P2099-0612.

74. Miles Lubin, J.A. Julian Hall, Cosmin G. Petra, Mihai Anitescu. "Parallel distributed-memory simplex for large-scale stochastic LP problems". Computational Optimization and Applications vol. 55, no. 3, pp. 571-596, 2013. Also, Preprint ANL/MCS-P2075-0412. Awarded Best COAP Paper Award for 2013, out of 92 eligible papers.

75. Michael L. Stein, Jie Chen and Mihai Anitescu, "STOCHASTIC APPROXIMATION OF SCORE FUNCTIONS FOR GAUSSIAN PROCESSES",Annals of Applied Statistics, Volume 7, Number 2 (2013), 1162-1191. Preprint ANL/MCSP-2091-0512.

76. A.Tasora, M.Anitescu, S.Negrini and D.Negrut. "A Compliant Visco-Plastic Particle Contact Model based on Differential Variational Inequalities". International Journal of Nonlinear Mechanics.; Volume 53, July 2013, Pages 2–12. Preprint ANL/MCS-P4013-0113.

77. Alessandro Tasora and Mihai Anitescu. "A complementarity-based rolling friction model for rigid contacts", Preprint ANL/MCS–P3020-0812.Meccanica. September 2013, Volume 48, Issue 7, pp 1643-1659. Also Preprint ANL/MCS–P3020-0812.

78. Hayes F. Stripling, Mihai Anitescu, and Marvin L. Adams." A Generalized Adjoint Framework for Sensitivity and Global Error Estimation in Time-Dependent Nuclear Reactor Simulations". Annals of Nuclear Energy (Special issue on Nuclear Reactor Safety Simulation and Uncertainty Analysis)Volume 52, February 2013, Pages 47-58. Also Preprint ANL/MCS-P1963-1011.

79. Emil M Constantinescu and Mihai Anitescu, "Physics-Based Covariance Models For Gaussian Processes With Multiple Outputs". International Journal on Uncertainty Quantification, Volume 3, No 1, pp 47-71, 2013. Preprint ANL/MCS-P1915-0711.

80. Michael Stein, Jie Chen, and Mihai Anitescu. "Difference Filter Preconditioning For Large Covariance Matrices". SIAM Journal on Matrix Analysis and Applications Vol. 33, No. 1, pp. 52-72, 2012. Preprint ANL/MCS-1888-0511.

81. Mihai Anitescu, Jie Chen, and Lei Wang, "A Matrix-Free Approach For Solving The Gaussian Process Maximum Likelihood Problem". SIAM Journal on Scientific Computing 2012, Vol. 34, No. 1, pp. A240-A262. Preprint ANL/MCS-P1857-0311.

82. Brian Lockwood and Mihai Anitescu. "Gradient-Enhanced Universal Kriging for Uncertainty Propagation". Nuclear Science and Engineering, Volume 170, Number 2, February 2012, Pages 168-195. Preprint ANL/MCS-P1808-1110.

83. Miles Lubin, Cosmin Petra, and Mihai Anitescu "On the parallel solution of dense saddle-point linear systems arising in stochastic programming". Optimization Methods and Software Volume 27, Issue 4-5, 2012. Preprint ANL/MCS-P1798-1010.

84. Cosmin Petra and Mihai Anitescu. "A preconditioning technique for Schur complement systems arising in stochastic optimization". Computational Optimization and Applications, Volume 52, Number 2 (2012), 315-344. Preprint ANL/MCS-P1748-0510.

85. Pointer, W. David, Mihai Anitescu, Thomas Fanning, Robert M. Ferencz, Paul Fischer, M. A. Smith, and T. Tautges. "SHARP reactor performance and safety analysis simulation suite." Transactions of the American Nuclear Society (2012).

86. Roderick, O, M Anitescu, and Z Wang. "Reduced Order Approximations in Uncertainty Analysis of Nuclear Engineering Applications." Transactions of the American Nuclear Society 106, 437-438 (2012).

87. Yiou Li, Mihai Anitescu, Oleg Roderick and Fred Hickernell, "ORTHOGONAL BASES FOR POLYNOMIAL REGRESSION WITH DERIVATIVE INFORMATION IN UNCERTAINTY QUANTIFICATION". International Journal for Uncertainty Quantification, 1 (4): 297–320 (2011). Also, Preprint ANL/MCS P1806-1110.

88. Oleg Roderick, Zhu Wang, Mihai Anitescu, "Dimensionality Reduction for Uncertainty Quantification of Nuclear Engineering Models". Trans. Am. Nucl. Soc.; Journal Volume: 104; Journal Issue: 2011. Preprint ANL/MCS-P1832-0111.

89. Brian A. Lockwood and Mihai Anitescu. "Gradient-Enhanced Universal Kriging for Uncertainty Propagation in Nuclear Engineering". Trans. Am. Nucl. Soc.; Journal Volume: 104; Journal Issue: 2011. Preprint ANL/MCS-P1833-0111.

90. Emil Constantinescu, Victor Zavala, Matthew Rocklin, Sangmin Lee, and Mihai Anitescu. "A Computational Framework for Uncertainty Quantification and Stochastic Optimization in Unit Commitment with Wind Power Generation". IEEE Transactions on Power Systems, 26(1), pp. 431–441 (2011). Also, Preprint ANL/MCS-P1681-1009.

91. Jie Chen, Mihai Anitescu and Yousef Saad. "Computing F(a)b Via Least Squares Polynomial Approximations". SIAM Journal on Scientific Computing, 33, pp. 195–222 (2011).

92. Alessandro Tasora and Mihai Anitescu, "A matrix-free cone complementarity approach for solving large-scale,nonsmooth, rigid body dynamics", Computer Methods in Applied Mechanics and Engineering, 200, 5-8, (2011), pp. 439–453. Also, Preprint ANL/MCS-P1692-1109.

93. Anitescu, Mihai. "Comments on: Algorithms for linear programming with linear complementarity constraints." TOP (2011). doi:10.1007/s11750-011-0229-1.

94. Mazhar, Hammad, Toby Heyn, Alessandro Tasora, Mihai Anitescu, and Dan Negruţ. "Million Body Simulations of Granular Dynamics on the GPU." 2011.

95. Victor Zavala and Mihai Anitescu. " On-Line Nonlinear Programming as a Generalized Equation", SIAM J. Control Optim., 48(8), pp. 5444–5467 (2010). Preprint ANL/MCS-P1641-0609.

96. Mihai Anitescu and Sanghyun Park."Short Communication: A Linear Assignment Approach for the Least-Squares Protein Morphing Problem Short Communication: A linear programming approach for the least-squares protein morphing problem". Mathematical Programming. Volume 125, Number 1, 195-203 (2010)

97. M Anitescu, A. Tasora. "An iterative approach for cone complementarity problems for nonsmooth dynamics". Preprint ANL/MCS-P1413-0507. Computational Optimization and Applications. Volume 47, Number 2, 207-235, (2010).

98. Mihai Alexe, Oleg Roderick, Mihai Anitescu, Jean Utke, Thomas Fanning, and Paul Hovland. "Using Automatic Differentiation in Sensitivity Analysis of Nuclear Simulation Models".Transactions of American Nuclear Society, volume 102, pages 235-237, 2010.

99. Alessandro Tasora and Mihai Anitescu." A Convex Complementarity Approach for Simulating Large Granular Flow". Journal of Computational Nonlinear Dynamics, Volume 5(3), Pages: 031004-, 2010.

100. D. Stewart and M. Anitescu. "Optimal Control of Systems with Discontinuous Differential Equations". Numerische Mathematik, Volume 114(4), Pages. 653–695, 2010.

101. Oleg Roderick, Mihai Anitescu, and Paul Fischer. "Stochastic finite element approaches using derivative information for uncertainty quantification" Nuclear Science and Engineering, Volume 164 (2), Pages 122–139, 2010.

102. Zavala, Victor M., and Mihai Anitescu. "Real-Time Nonlinear Optimization as a Generalized Equation." SIAM Journal on Control and Optimization (2010). doi:10.1137/090762634.

103. Oleg Roderick, Mihai Anitescu, Paul Fischer, Won-Sik Yang. "Stochastic Finite-Element Approach in Nuclear Reactor Uncertainty Quantification". Transactions of American Nuclear Society, volume 100, pages 317-318, 2009.

104. Victor Zavala, Emil Constantinescu, Theodore Krause, and Mihai Anitescu." Journal of Process Control, Volume 19, Issue 10, Pages 1725-1736, December 2009.

105. Cosmin Petra, Bogdan Gavrea, Mihai Anitescu, and Florian Potra. "A computational study of the use of an optimization-based method for simulating large multibody systems ". Optimization Methods and Software. Volume 24(6), Pages 871–894, 2009.

106. Kyle Schmitt, Mihai Anitescu and Dan Negrut. "Efficient Sampling of Dynamical Systems with Spatial Uncertainty". International Journal for Numerical Methods in Engineering (IJNME).Volume 80(5), Pages 537-564, 2009.

107. M. Anitescu. Spectral Stochastic Finite Element Methods for Parametric Constrained Optimization Problems. SIAM Journal of Numerical Analysis 47, Issue 3, pp. 1739-1759 (2009).

108. Xiaoyan Zeng, Mihai Anitescu, Candido Pereira, and Monica Regalbuto. "A Framework for Chemical Plant Safety Assessment Under Uncertainty ". Studies in Informatics and Control. Volume 18(1), Pages 5-20, (2009)

109. Jaydeep Bardhan, Matt Knepley and Mihai Anitescu. "Bounding the Electrostatic Free Energies Associated with Linear Continuum Models of Molecular Solvation". J. Chem. Phys. 130, 104108, 2009.

110. M. Anitescu, D. Negrut, A. El-Azab, and P. Zapol, A note on the regularity of reduced models obtained by nonlocal quasicontinuum-like approaches. Mathematical Programming. Volume 118, Number 2, Pages 207-236, 2009.

111. Schmitt, Kyle P., Mihai Anitescu, and Dan Negrut. "Efficient sampling for spatial uncertainty quantification in multibody system dynamics applications." International Journal for Numerical Methods in Engineering (2009). doi:10.1002/nme.2649.

112. Zavala, Victor M., Emil M. Constantinescu, Theodore Krause, and Mihai Anitescu. "On-line economic optimization of energy systems using weather forecast information." Journal of Process Control (2009). doi:10.1016/j.jprocont.2009.07.004.

113. Alessandro Tasora, Dan Negrut and Mihai Anitescu. "Large-Scale Parallel Multibody Dynamics with Frictional Contact on the Graphical Processing Unit ". Proc. IMechE, Part K: J. Multi-body Dynamics, 222(K4), 315-326, 2008.

114. B. Gavrea, M. Anitescu, and F.A. Potra. Convergence of a Class of Semi-Implicit Time-Stepping schemes for Nonsmooth Rigid Multibody Dynamics.SIAM J. Optim 19(2), pp. 969-1001 (2008).

115. D. Negrut, M. Anitescu, A. El-Azab, and P. Zapol, "Quasicontinuum-like reduction of DFT calculations of nanostructures", Journal of Nanoscience and Nanostructures, Volume 8, Pages 3729–3740, 2008.

116. Mihai Anitescu, Paul Tseng, Stephen J. Wright. Elastic-Mode Algorithms for Mathematical Programs with Equilibrium Constraints: Global Convergence and Stationarity Properties. Mathematical Programming, 110(2) July, pp 337-371 2007.

117. Anitescu, M. Hovland, P. Palmiotti, G. Yang, W. S. "An Overview of Automatic Differentiation Tools and Techniques for Nuclear Reactor Applications", Transactions- American Nuclear Society, vol. 96, pages 538-539, 2007.

118. Anitescu, M. Hovland, P. Palmiotti, G. Yang, W. S. "Randomized Quasi Monte Carlo Sampling Techniques in Nuclear Reactor Uncertainty Assessment", Transactions- American Nuclear Society, Vol. 96, pages 526-527, 2007.

119. Mihai Anitescu, William J. Layton. Sensitivities in Large Eddy Simulation and Improved Estimates of Turbulent Flow Functionals. SIAM Journal of Scientific Computing, Volume 29 Issue 4, Pages 1650-1667, 2007.

120. Tasora, Alessandro, and Mihai Anitescu. "A fast contraction mapping for solving multibody systems." PAMM (2007). doi:10.1002/pamm.200700020.

121. M. Anitescu. Optimization-based simulation of nonsmooth dynamics. Mathematical Programming, series A, 105, pp 113–143, 2006.

122. F.A. Potra, M. Anitescu, B. Gavrea and J. Trinkle. "A linearly implicit trapezoidal method for stiff multibody dynamics with for multi-rigid-body dynamics with contact and friction", International Journal for Numerical Methods in Engineering, 66(7), pp 1079-1124, 2006.

123. G. Aliberti, G. Palmiotti, M. Salvatores, T.K. Kim, T.A. Taiwo, M. Anitescu, I. Kodeli, E. Sartori, J.C. Bosq, J. Tommasi Nuclear Data Sensitivity, Uncertainty and Target Accuracy Assessment for Future Nuclear Systems. Annals of Nuclear Energy 33(8), pp 700-733, 2006.

124. Potra, Florian A., Mihai Anitescu, Bogdan Gavrea, and Jeff Trinkle. "A linearly implicit trapezoidal method for integrating stiff multibody dynamics with contact, joints, and friction." International Journal for Numerical Methods in Engineering (2006). doi:10.1002/nme.1582.

125. M. Anitescu. On Solving Mathematical Programs with Complementarity Constraints as Nonlinear Programs. SIAM Journal on Optimization, 15(4), pp. 1203–1236, 2005.

126. M. Anitescu. Global Convergence of an Elastic Mode Approach for A Class of Mathematical Programs with Complementarity Constraints. SIAM Journal on Optimization, 16(1), pp 120–145, 2005.

127. Anitescu, Mihai. "On Using the Elastic Mode in Nonlinear Programming Approaches to Mathematical Programs with Complementarity Constraints." SIAM Journal on Optimization (2005). doi:10.1137/s1052623402401221.

128. M. Anitescu and G.D. Hart. A constraint-stabilized time-stepping approach for rigid multibody dynamics with joints, contact and friction. International Journal for Numerical Methods in Engineering, 60(14), 2335-2371, 2004.

129. M. Anitescu and G.D. Hart. A Fixed-Point Iteration Approach for Multibody Dynamics with Contact and Small Friction. Math Programming B. 101(1), 3–32, 2004.

130. M. Anitescu, Faranak Pahlevani and William J. Layton. "Implicit for Local Effects and Explicit for Local Effects Is Unconditionally Stable". Electronic Transactions in Numerical Analysis, 18, 174–187, 2004.

131. M. Anitescu and G. D. Hart. Solving nonconvex problems of multibody dynamics with contact and small friction by sequential convex relaxation. Mechanics Based Design of Machines and Structures 31(3), 335-356, 2003.

132. M. Anitescu. On the rate of convergence of Sequential Quadratic Programming with nondifferentiable exact penalty function in the presence of constraint degeneracy. Mathematical Programming, 92 (2002), 359-386.

133. M. Anitescu and F. A. Potra. Time-Stepping Schemes for Stiff Multi-Rigid-Body Dynamics with Contact and Friction. International Journal for Numerical Methods in Engineering, 55 (2002), 753–784.

134. M. Anitescu. A Superlinearly Convergent Sequential Quadratically Constrained Quadratic Programming Algorithm for Degenerate Nonlinear Programming, SIAM Journal of Optimization 12 (2002), 949-978.

135. M.K. Mihcak, P. Moulin, M. Anitescu and K. Ramchandran, Rate-Distortion Optimal Subband Coding Without Perfect-Reconstruction Constraints, IEEE Transactions in Signal Processing, 49(3): 542-557,2001.

136. P.Moulin, M. Anitescu and K. Ramchandran, Theory of Rate–Distortion–Optimal, Constrained Filter Banks — Application to IIR and FIR Biorthogonal Designs, IEEE Transactions in Signal Processing, 48(4):1120–1132, 2000.

137. M. Anitescu, Degenerate Nonlinear Programming with a Quadratic Growth Condition, SIAM Journal of Optimization, 10(4): 1116-1135, 2000.

138. M. Anitescu, F.A.Potra and D. Stewart, Time-Stepping for Three-Dimensional Rigid-Body Dynamics, Computer Methods in Applied Mechanics and Engineering, 177, pp. 183–197, 1999.

139. M. Anitescu, G. Lesaja and F. A. Potra, Equivalence between different formulations of the linear complementarity problems, Optimization Methods and Software 7 (3-4), pp. 265-290, 1997.

140. M. Anitescu, G. Lesaja and F. A. Potra, An infeasible-interior-point predictor-corrector algorithm for the P_*-Geometric LCP, Applied Mathematics and Optimization, 36 (2), pp. 203-228, 1997.

141. P. Moulin, M. Anitescu, K. O. Kortanek and F. A. Potra, The Role of Linear Semi-Infinite Programming in Signal-Adapted QMF Bank Design, IEEE Transactions of Signal Processing, 45 (9), pp. 2160-2174, 1997.

142. M. Anitescu and F.A.Potra, Formulating dynamic multi-rigid-body contact problems with friction as solvable Linear Complementarity Problems Nonlinear Dynamics, 14, pp. 231–247, 1997.

143. M. Anitescu, J. F. Cremer and F. A. Potra, Formulating 3D Contact Dynamics Problems, Mechanics of Structures and Machines, 24(4), November 1996.

144. M. Anitescu, D. I. Coroian, M. Z. Nashed and F. A. Potra, Outer Inverses and Multi-Body System Simulation, Numerical Functional Analysis and Optimization, 17 (7-8), pp 661-678, 1996.

BOOK CHAPTERS

1. Pacaud, François, Michel Schanen, Daniel Adrian Maldonado, Alexis Montoison, Valentin Churavy, Julian Samaroo, and Mihai Anitescu. "Batched Second-Order Adjoint Sensitivity for Reduced Space Methods." Proceedings of the 2022 SIAM Conference on Parallel Processing for Scientific Computing (PP) (2022). doi:10.1137/1.9781611977141.6.

2. Alesandro Tasora, Dan Negrut, and Mihai Anitescu. GPU-based Parallel Computing for the Simulation of Complex Multibody Systems with Unilateral and Bilateral Constraints: An Overview. To appear in "Computational Methods in Applied Sciences", Springer, 2009. (Proceedings of ECCOMAS).

3. A. Tasora, M. Anitescu. " A Fast NCP Solver for Large Rigid-Body Problems with Contacts, Friction, and Joints." Springer book for ECCOMAS Thematic Conference in Multibody Dynamics, Editor, Carlo Botasso, Springer, Berlin, 2008, pp 45-55.

4. M. Anitescu and F.A. Potra "On integrating stiff rigid multibody dynamics with contact and friction", in Contact Mechanics. Proceedings of the 3rd Contact Mechanics International Symposium, Kluwer Academic Publishers, Dordrecht, Netherlands, 2002, 83–90.

5. M. Anitescu, J. F. Cremer and F. A. Potra, Properties of Complementarity Formulations for Contact Problems with Friction, in Complementarity and Variational Problems: State of the Art, Michael C. Ferris and Jong-Shi Pang, Editors, SIAM Publications, Philadelphia, 1997, pp. 12-21.

6. Dan Negrut, Alessandro Tasora, Mihai Anitescu, Hammad Mazhar, Toby Heyn, Arman Pazouki. Solving Large Multibody Dynamics Problems on the GPU. To appear in GPU Gems 4. Preprint ANL/MCS-1777-0710. Pdf Version.

7. Victor M. Zavala, Mihai Anitescu and Theodore Krause. "On the Optimal On-Line Management of Photovoltaic-Hydrogen Hybrid Energy System". To appear in the Proceedings of the 10th International Symposium on Process Systems Engineering - PSE2009, Elsevier, Amsterdam.

REFEREED PROCEEDINGS PAPERS

1. Athalye, S, MS Ramadan, M Anitescu, and KG Vamvoudakis. "Data-Conforming Model-Free Stochastic Reinforcement Learning." 2026 American Control Conference (ACC), 3819-3824 (2026).

2. Wang, Y, ZW Anderson, V Rao, M Anitescu, S Rosenkranz, and R Osborn. "Symmetry-constrained machine learning on 3D-ΔPDF to identify short range correlations." APS Global Physics Summit 2026 (2026).

3. Shin, Sungho, Vishwas Rao, Michel Schanen, D. Adrian Maldonado, and Mihai Anitescu. "Scalable Multi-Period AC Optimal Power Flow Utilizing GPUs with High Memory Capacities." 2024 Open Source Modelling and Simulation of Energy Systems (OSMSES) (2024). doi:10.1109/osmses62085.2024.10668970.

4. Shin, Sungho, François Pacaud, Emil Contantinescu, and Mihai Anitescu. "Constrained Policy Optimization for Stochastic Optimal Control under Nonstationary Uncertainties*." 2023 American Control Conference (ACC) (2023). doi:10.23919/acc55779.2023.10156553.

5. Cole, David, Sungho Shin, François Pacaud, Victor M. Zavala, and Mihai Anitescu. "Exploiting GPU/SIMD Architectures for Solving Linear-Quadratic MPC Problems*." 2023 American Control Conference (ACC) (2023). doi:10.23919/acc55779.2023.10155791.

6. Luan, Lele, Nesar Ramachandra, Sandipp Krishnan Ravi, Anindya Bhaduri, Piyush Pandita, Prasanna Balaprakash, Mihai Anitescu, Changjie Sun, and Liping Wang. "Scalable Probabilistic Modeling and Machine Learning With Dimensionality Reduction for Expensive High-Dimensional Problems." Volume 2: 43rd Computers and Information in Engineering Conference (CIE) (2023). doi:10.1115/detc2023-110704.

7. Shin, S, F Pacaud, and M Anitescu. "Nonlinear Optimization on Exascale Computing Architectures and GPUs." 2023 AIChE Annual Meeting (2023).

8. Lei, Shaolin, Daniel Adrian Maldonado, Emil M. Constantinescu, Junbo Zhao, Somayeh Yarahmadi, Lamine Mili, and Mihai Anitescu. "A Novel Continuum Approximation to Power System Electromechanical Dynamics." 2022. doi:10.1109/isgt50606.2022.9817495.

9. Rao, Vishwas, Anirudh Subramanyam, Michel Schanen, Youngdae Kim, Ignas Satkauskas, and Mihai Anitescu. "Frequency Recovery in Power Grids using High-Performance Computing." Workshop Proceedings of the 51st International Conference on Parallel Processing (2022). doi:10.1145/3547276.3548632.

10. Upreti, P, M Krogstad, C Haley, M Anitescu, V Rao, and L Poudel. "Order-disorder vs Displacive structural phase transitions in (CaxSr1 - x)3Rh4Sn13." APS March Meeting Abstracts 2022, A63. 008 (2022).

11. Maldonado, Daniel Adrian, Michel Schanen, François Pacaud, and Mihai Anitescu. "Domain Decomposition Preconditioners for Unstructured Network Problems in Parallel Vector Architectures." 50th International Conference on Parallel Processing Workshop (2021). doi:10.1145/3458744.3473363.

12. Rao, Vishwas, Romit Maulik, Emil Constantinescu, and Mihai Anitescu. "A Machine-Learning-Based Importance Sampling Method to Compute Rare Event Probabilities." In International Conference on Computational Science, pp. 169-182. Springer, Cham, 2020.

13. Maldonado, Daniel Adrian, Vishwas Rao, Mihai Anitescu, and Vivak Patel. "Sequential Bayesian parameter estimation of stochastic dynamic load models." Electric Power Systems Research 189 (2020): 106606.

14. Kim, Youngdae, and Mihai Anitescu. "A Real-Time Optimization with Warm-Start of Multiperiod AC Optimal Power Flows." Electric Power Systems Research 189 (2020): 106721.

15. Shin, Sungho, Mihai Anitescu, and Victor M. Zavala. "Overlapping Schwarz Decomposition for Constrained Quadratic Programs." 2020 59th IEEE Conference on Decision and Control (CDC) (2020). doi:10.1109/cdc42340.2020.9304139.

16. Maldonado, Daniel Adrian, and Mihai Anitescu. "Variance-Based Sensitivity Analysis of the Composite Dynamic Load Model." 2020 IEEE Power & Energy Society General Meeting (PESGM) (2020). doi:10.1109/pesgm41954.2020.9281523.

17. Schanen, M, A Maldonado, F Pacaud, and M Anitescu. "ExaPF. jl: a power flow solver for GPUs." Proceedings of JuliaCon 1, 1 (2020).

18. Shin, S, M Anitescu, and VM Zavala. "Overlapping Schwarz: A New Decomposition Paradigm for Large-Scale Optimization." 2020 Virtual AIChE Annual Meeting (2020).

19. Kim, Cheolmin, Kibaek Kim, Prasanna Balaprakash, and Mihai Anitescu. "Graph convolutional neural networks for optimal load shedding under line contingency." In 2019 IEEE Power & Energy Society General Meeting (PESGM), pp. 1-5. IEEE, 2019.

20. Rao, Vishwas, Kibaek Kim, Michel Schanen, Daniel A. Maldonado, Cosmin Petra, and Mihai Anitescu. "A multiperiod optimization-based metric of grid resilience." In 2019 IEEE Power & Energy Society General Meeting (PESGM), pp. 1-5. IEEE, 2019.

21. Schanen, Michel, Daniel Adrian Maldonado, and Mihai Anitescu. "A Framework for Distributed Approximation of Moments with Higher-Order Derivatives Through Automatic Differentiation." In International Conference on Computational Science, pp. 251-260. Springer, Cham, 2019.

22. Shin, S, M Anitescu, and VM Zavala. "Overlapping Domain Decomposition Schemes for Graph-Structured Optimization Problems." 2019 AIChE Annual Meeting (2019).

23. Vivak Patel, Adrian Maldonado, and Mihai Anitescu. Semiparametric Estimation of Solar Generation. To appear in Proceedings of the 2018 IEEE PES General meeting.

24. Kibaek Kim, Mihai Anitescu, and Victor Zavala, An Asynchronous Decomposition Algorithm for Security Constrained Unit Commitment under Contingency Events. To appear in Proceedigs of PSCC 2018.

25. Mihai Anitescu, Francois Gilbert, Cosmin Petra, Michel Schanen. "Toward Multi Period AC-Based Contingency Constrained Optimal Power Flow at Large Scale". To appear in Proceedings of PSCC 2018.

26. Mihai Anitescu and Victor Zavala. "MPC as a DVI: Implications on Sampling Rates and Accuracy". To appear in Proceedings of CDC 2017.

27. D. Maldonado, V. Patel, M. Anitescu, A. Flueck."A Statistical Approach to Dynamic Load Modeling and Identification with High Frequency Measurements ". To appear in proceedings of IEEE PES 2017. Selected for Best Paper Presentation.

28. Michel Schanen, Oana Marin, Hong Zhang, Mihai Anitescu. Asynchronous Two-Level Checkpointing Scheme for Large-Scale Adjoints in the Spectral-Element Solver Nek5000. Preprint ANL/MCS-P5422-1015. In Proceedings of ICCS 2016

29. Liang, L, ZG Mei, B Ye, YS Kim, G Hofman, M Anitescu, and AM Yacout. "Multiscale simulation of microstructural evolution in irradiated U-Mo." European Research Reactor Conference (RRFM) 2016: Conference Proceedings, 77-85 (2016).

30. Qiang, F, C Petra, M Lubin, J Huchette, and M Anitescu. "On efficient Hessian computation using the edge pushing algorithm in Julia." Proceedings of the 7th International Conference on Algorithmic Differentiation (2016).

31. Liang, Linyun, Marius Stan, and Mihai Anitescu. "Phase Field Modeling of the Li2O2 Growth in a Porous Cathode in Li-Air Batteries with Organic Electrolytes." ECS Meeting Abstracts (2015). doi:10.1149/ma2015-01/2/531.

32. Mei, Z, L Liang, YS Kim, T Wiencek, G Hofman, M Anitescu, and AM Yacout. "Grain Growth and Bubble Evolution in U-Mo Alloy by Multiscale Simulations." International Meeting on RETER 2015-36th International Meeting on Reduced ... (2015).

33. S. Abhyankar, V. Rao, and M. Anitescu, "Dynamic Security Constrained Optimal Power Flow Using Finite Difference Sensistivities", To appear in Transactions of the IEEE Power & Energy Society General Meeting, 2014. Also Preprint ANL/MCS-P5041-1213.

34. Rosenkranz, S, J Chen, DJP Morris, SN Ancona, H Zheng, and JF Michell. "Quantitative analysis of the diffuse scattering from a bilayer manganite." APS March Meeting Abstracts 2014, L6. 003 (2014).

35. Roderick, O, Anitescu, M. Use of Multi-Fidelity Training Data in Uncertainty Analysis of Nuclear Engineering Applications. To appear in Proceedings of the American Nuclear Society Meeting 2013; Atlanta, Georgia Preprint ANL/MCS-P4074-0513.

36. A. Mametjanov, Norris, B, Zeng, X, Drewniak, BA, Utke, J, Anitescu, M, Hovland, PD. Applying Automatic Differentiation to the Community Land Model, in 6th International Conference on Automatic Differentiation (AD2012), Fort Collins, CO, 2012.ANL/MCS-P1993-0112.

37. Brian Lockwood, Mihai Anitescu; Dimitri Mavriplis. Mixed Aleatory/Epistemic Uncertainty Quantification for Hypersonic Flows via Gradient-Based Optimization and Surrogate Models. AIAA-2012-1254 50th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, Nashville, Tennessee, Jan. 9-12, 2012.

38. Heyn, Toby, Dan Negrut, Mihai Anitescu, Alessandro Tasora, and David Lamb. "On the Numerical Solution of Many-Body Contact Dynamics Problems Formulated as Complementarity Problems." Volume 6: 1st Biennial International Conference on Dynamics for Design; 14th International Conference on Advanced Vehicle Technologies (2012). doi:10.1115/detc2012-71148.

39. Collins, William D., Mihai Anitescu, Daniel Bader, Bert Debusschere, James Gattiker, Paul Hovland, Gunnar Þór Jóhannesson, Ge Lin, Damian Lucas, Habib N. Najm, Yun Qian, and Laura Swiler. "The Role of Uncertainty Quantification in Climate Science for a Sustainable Energy Future." AGUFM (2012).

40. Miles Lubin, Cosmin G. Petra, Mihai Anitescu, and Victor Zavala. "Scalable Stochastic Optimization of Complex Energy Systems". To appear in proceedings of Supercomputing 2011 (acceptance rate: 21%).

41. LeiWang, Jungho Lee, Mihai Anitescu, Anter El Azab, Lois Curfman McInnes, Todd Munson, and Barry Smith." A Differential Variational Inequality Approach for the Simulation of Heterogeneous Materials". To appear in proceedings of the 2011 SCIDAC Conference.

42. Zavala, Victor M., and Mihai Anitescu. "New insights into the dynamic stability of wholesale electricity markets." 2011 IEEE International Symposium on Computer-Aided Control System Design (CACSD) (2011). doi:10.1109/cacsd.2011.6044562.

43. Drewniak, Beth, Xiaoyan Zeng, A. Mametjanov, Mihai Anitescu, Barnaby Norris, and V. R. Kotamarthi. "Parameter Sensitivity Evaluation of the CLM-Crop model." AGUFM (2011).

44. Zavala, V.M. and Anitescu, M. "Achieving higher frequencies in large-scale nonlinear model predictive control", 49th IEEE Conference on Decision and Control, Atlanta, GA, USA on December 15-17, 2010.

45. Zavala, V. M.; Wang, J.; Leyffer, S.; Constantinescu, E.M.; Anitescu, M. and Conzelmann, G. "Proactive Energy Management for Next-Generation Building System"s. SimBuild, 2010

46. Anitescu, Mihai, Emil M. Constantinescu, and Víctor M. Zavala. "Optimal Management and Design of Energy Systems under Atmospheric Uncertainty." AGUFM (2010).

47. Tasora, Alessandro, Dan Negruţ, Mihai Anitescu, Hammad Mazhar, and Toby Heyn. "Simulation of Massive Multibody Systems." Digital Library (University of West Bohemia) (2010).

48. Oleg Roderick, Mihai Anitescu, Paul Fischer, Won-Sik Yang. Polynomial regression with derivative information in nuclear reactoruncertainty quantification. To appear in Proceedings of the 2009 Fast Reactor Meeting (Kyoto).

49. Victor M. Zavala, Emil M. Constantinescu, and Mihai Anitescu. Economic Impacts of Advanced Weather Forecasting in Energy System Operations.To appear in Proceedings of the 2009 IEEE-PES conference.

50. Heyn, T., H. Mazhar, A. Tasora, M. Anitescu, and D. Negrut,"A parallel algorithm for solving complex multibody problems with stream processors". 2009 ASME IDETC conference, paepr DETC2009-86478.

51. Heyn, T., H. Mazhar, A. Tasora, M. Anitescu, and D. Negrut,"A parallel algorithm for solving complex multibody problems with stream processors", in ECCOMAS Multibody Dynamics Conference, J. Franczek, Editor. 2009: Warsaw, Poland.

52. Tasora, A, D Negrut, and M Anitescu. "Parallel Large Scale Granular Flow Dynamics on the GPU." Proceedings of ECCOMAS 2009, 1-16 (2009).

53. Alessandro Tasora, Dan Negrut, Mihai Anitescu. Large-scale parallel multibody dynamics with frictional contact on the GPU. To appear in Proceedings of the ASME 2008 Dynamics and Control Conference, DSCC 2008. Preprint ANL/MCS -P1487-0308

54. Alessandro Tasora, Dan Negrut, Mihai Anitescu. "A GPU-based implementation of a cone convex complementarity approach for simulating rigid body dynamics with frictional contact". To appear in proceedings of the 2008 ASME Congress, paper IMECE-66766. Preprint ANL/MCS-P1545-0908

55. Kyle Schmitt, Justin Madsen, Mihai Anitescu, Dan Negrut. " A Gaussian process based approach for handling uncertainty in vehicle dynamics simulation". Preprint ANL/MCS-P1505-0608. Pdf version. To appear in the Proceedings of the 2008 ASME Congress, paper number IMECE-66664.

56. M. Anitescu, G. Palmiotti, Won-Sik Yang and Monika Neda. "Stochastic Finite-Element Approximation Of The Parametric Dependence Of Eigenvalue Problem Solution". Proceedings of the Mathematics, Computation and Supercomputing in Nuclear Applications, Monterey, 2007.

57. Dan Negrut, Mihai Anitescu, Anter El-Azab, Steve Benson, Peter Zapol, Emil Constantinescu, and Toby Heyn, A Real-Space Parallel Optimization Model Reduction Approach for Electronic Structure Computation in Large Nanostructures Using Orbital-Free Density Functional Theory, In Proceedings of the 2006 ASME International Mechanical Engineering Congress and Exposition, paper no. IMECE 2006-15740, 2006.

58. M. Anitescu. "Mathematical Programs with Equilibrium Constraints ans Applications to Control". Argonne National Laboratory, Mathematics and Computer Science Division. Proceedings of the 9th International Conference on Control Systems and Computer Science, Bucharest, 2005.

59. D. Negrut, M. Anitescu, T. Munson, and P. Zapol, "Simulating nanoscale processes in solids using DFT and the quasicontinuum method", in Proceedings of IMECE 2005, ASME International Mechanical Engineering Congress and Exposition, paper no. IMECE 2005-81755, 2005.

60. J. Peng, M. Anitescu and Srinivas Akella."Mathematical Programs with Complementarity Constraints for Multiple Robot Coordination with Contact and Friction". Proceedings of the 2004 International Conference in Robotics and Automation.

61. Peng, Jufeng, M. Anitescu, and S. Akella. "Optimal control of multiple robot systems with friction using mpcc." IEEE International Conference on Robotics and Automation, 2004. Proceedings. ICRA '04. 2004 (2004). doi:10.1109/robot.2004.1302547.

62. M. Anitescu, A. Miller and G. Hart. "Constraint stabilization for time-stepping approaches for rigid multibody dynamics with joints, contact and friction. Proceedings of ASME 2003 Design Engineering Technical Conferences, Chicago, Illinois, September 2003, Paper DETC2003/VIB-48432.

63. M. Anitescu. "A Fixed Time Stepping Approach for Multibody Dynamics With Contact and Friction". Proceedings of the 2003 IEEE International Symposium in Intelligent Robots and Systems, Las Vegas, Nevada, October 2003, 3725–3731.

64. G.D. Hart and Mihai Anitescu. "A Hard-Constraint Time-Stepping Approach for Rigid Multibody Dynamics with Joints, Contact, and Friction". Proceedings of the ACM 2003 Tapia Conference for Diversity in Computing, Atlanta, Georgia, October 2003, 34–40.

65. P.Moulin, M.Anitescu and K. Ramchandran, Asymptotic performance of subband coders using constrained, signal-adapted FIR filter banks, Proceedings of the Conference on Information Sciences and Systems, Princeton, March 1998.

66. K.Mihcak, P.Moulin, K.Ramchandran and M. Anitescu, Some Fundamental Properties of MMSE filter Banks. Proceedings of the 1998 IEEE-SP International Symposia on Time-Frequency and Time-Scale analysis, Pittsburgh, October 1998.

67. P. Moulin, M. Anitescu, K. O. Kortanek and F. A. Potra, Design of Signal-Adapted FIR Paraunitary Filter Banks, Proceedings of International IEEE Conference on Acoustics, Speech and Signal Processing, Atlanta, 1996 Volume 3, pp. 1519-1522.

68. M. Anitescu and J.F. Cremer, Linear Complementarity Problems in Frictionless Contact Dynamics Proceedings of the 10th International Conference on Control Systems and Computer Science, Bucharest, Vol.1, pp 87-94, May 1995.

69. M. Anitescu, A Generalization of Kharitonov's Theorem, Proceedings of the 9th International Conference on Control Systems and Computer Science, Bucharest, Vol. 1, pp. 89-95, May 1993.

70. D. A. Maldonado, Michel Schanen, and Mihai Anitescu. Uncertainty Propagation in Power System Dynamics with the Method of Moments. Selected as Best Conference Paper (4 such selections out of 1300+ submissions)

71. Victor Zavala and Mihai Anitescu. Stability and Robustness of Wholesale Electricity Markets. Proceedings of the 9th International Conference on Control Systems and Computer Science, Bucharest, Romania.

TECHNICAL REPORTS

1. Daniel, Claus, USDOE Office of Science (SC), USDOE National Nuclear Security Administration (NNSA), Jess Gehin, Kirsten Laurin-Kovitz, Bryan Morreale, Rick Stevens, William Tumas, Mihai Anitescu, Alec Poczatek, Andrew Siegel, Sibendu Som, Richard Vilim, Ray Grout, Benjamin Kroposki, Meng Yue, Kelly Rose, Ahmad Al Rashdan, Christopher Ritter, Prasanna Balaprakash, Prashant Jain, Teja Kuruganti, MA Piette, Tianzhen Hong, Court Corley, Robert Ralló, John Grosh, Brian Van Essen, Matt Reno, Hari Viswanathan, Frank Alexander, and Emily Dietrich. "Advanced Research Directions on AI for Energy." 2024. doi:10.2172/2340139.

2. Buluç, Aydın, Tamara G. Kolda, Stefan M. Wild, Mihai Anitescu, Anthony M. DeGennaro, John Jakeman, Chandrika Kamath, Ramakrishnan Kannan, Miles E. Lopes, Per-Gunnar Martinsson, Kary Myers, Jelani Nelson, Juan M. Restrepo, C. Seshadri, Draguna Vrabie, Brendt Wohlberg, Stephen Wright, Chao Yang, and Peter H. Zwart. "Randomized Algorithms for Scientific Computing (RASC)." 2021. doi:10.2172/1807223.

3. States), Argonne National Lab. (ANL), Argonne, IL (United, Shrirang Abhyankar, Mihai Anitescu, Emil Constantinescu, and Hong Zhang. "Efficient Adjoint Computation of Hybrid Systems of Differential Algebraic Equations with Applications in Power Systems." 2016. doi:10.2172/1245175.

4. Petra, CG, V Zavala, E Nino-Ruiz, and M Anitescu. "Economic impacts of wind covariance estimation on power grid operations." Preprint ANL/MCS-P5M8-0614 (2014).

5. Chen, J, L Wang, and M Anitescu. "A parallel tree code for computing matrix-vector products with the Matérn kernel." Techni-cal Report no. ANL/MCS-P50, 15-09 (2013).

6. Alexander, Francis J., Mihai Anitescu, Jeffery W. Bell, David T. Brown, Malcolm Ferris, Mitchell Luskin, Sanjay Mehrotra, Bryan Moser, A. Pinar, Alexandre M. Tartakovsky, Karen Willcox, Stephen J. Wright, and Víctor M. Zavala. "A Multifaceted Mathematical Approach for Complex Systems." 2012. doi:10.2172/1093585.

7. (SC), USDOE Office of Science, Oleg Roderick, Mihai Anitescu, and Jean Utke. "Derivative-based uncertainty quantification: automatic differentiation tools for SAS." 2012. doi:10.2172/1055419.

8. (SC), USDOE Office of Science, O. Roderick, and Mihai Anitescu. "Intrusive analysis for NEK5000: development of intrusive uncertainty quantification for high-dimensional, high-fidelity codes." 2012. doi:10.2172/1056194.

9. Heyn, T, M Anitescu, A Tasora, and D Negrut. "A comparison of several solvers for bound constraint quadratic programming within the context of frictionless multibody dynamics." Technical Report TR-2012-02, Simulation-Based Engineering Laboratory ... (2012).
10. Mihai Anitescu and Oleg Roderick "Verification, Validation, Uncertainty Quantification Tasks for the SHARP Reactor Performance and Safety Code". Technical Memo ANL/MCS--TM--318, 2011 (Refereed Technically).
11. Mihai Anitescu and Oleg Roderick. "Intrusive Analysis and Uncertainty Quantification for Nuclear Engineering Applications". Technical Memo ANL/MCS–TM–317, 2011 (Refereed Technically).
12. Anitescu, M, and C Petra. "Higher-order confidence intervals for stochastic programming using bootstrapping." Tech. Rep. ANL/MCS-P1964-1011, Mathematics and Computer Science Division ... (2011).
13. Zavala, VM, M Anitescu, A Kannan, and C Petr. "On the Convergence of Day-Ahead and Real-Time Electricity Markets." FERC.[Online]. Available at: http://www. ferc. gov/CalendarFiles ... (2011).
14. Victor Zavala, Emil Constantinescu, Theodore Krause and Mihai Anitescu. "Exploiting Weather Forecast in integrated energy systems".Technical MemoANL/MCS-TM-305. March 2009.
15. E.M. Constantinescu, V.M. Zavala, M. Rocklin, S. Lee, and M. Anitescu. "Unit Commitment with Wind Power Generation: Integrating Wind Forecast Uncertainty and Stochastic Programming". Memo ANL/MCS-TM309. August 2009.
16. Mihai Alexe, Oleg Roderick, Jean Utke, Mihai Anitescu, Paul Hovland, and Thomas Fanning. "Automatic Differentiation of Codes in Nuclear Engineering Application". Memo ANL/MCS-TM310, November 2009
17. Zavala, Víctor M., Emil M. Constantinescu, Theodore R. Krause, and Mihai Anitescu. "Weather forecast-based optimization of integrated energy systems." 2009. doi:10.2172/951585.
18. Mihai Anitescu and John Birge." Convergence of stochastic average approximation for stochastic optimization problems with mixed expectation and per-scenario constraints". Preprint ANL/MCS 1562-1108. November 2008.
19. G. Palmiotti, J. Cahalan, P. Pfeiffer, T. Sofu, T. Taiwo, T. Wei, A. Yacout, W. Yang, A. Siegel, Z. Insepov, M. Anitescu,P. Hovland, C. Pereira, M. Regalbuto, J. Copple, M. Williamson. " Requirements for Advanced Simulation of Nuclear Reactor and Chemical Separation Plants". Technical Report ANL-AFCI-168, December 2006.
20. Dan Negrut, Mihai Anitescu, Todd Munson, Peter Zapol, Density Functional Theory-Based Nanostructure Investigation: Theoretical Considerations. Preprint number: ANL/MCS-P1252-0505, Mathematics and Computer Science Division, Argonne National Laboratory, May 2005.
21. M. Anitescu, Nonlinear Programs with Unbounded Lagrange Multiplier Sets, Preprint ANL/MCS-P793-0200, Argonne National Laboratory, Argonne, Ill., 2000.
22. M. Anitescu and R. Serban, "A Sparse Superlinearly Convergent SQP with Applications to Two-dimensional Shape Optimization," Preprint ANL/MCS-P706-0198, 1998.
23. Gary Hart and Mihai Anitescu. "An O(m+n) measure of penetration depth between convex polyhedral bodies for rigid multibody dynamics". Preprint ANL/MCS-P1753-0510.

PREPRINTS AND UNPUBLISHED MANUSCRIPTS

1. Shin, Sungho, Michel Schanen, François Pacaud, Alexis Montoison, and Mihai Anitescu. "ExaModels.jl: an Algebraic Modeling System for Nonlinear Programming on GPUs." arXiv (Cornell University) (2026). doi:10.48550/arxiv.2608.16265.
2. Huang, Boxun, David Y. Shu, Michel Schanen, Mihai Anitescu, R Gracy Gandhi, and Sungho Shin. "Harnessing GPU Acceleration in Large-Scale Process Optimization." arXiv (Cornell University) (2026). doi:10.48550/arxiv.2607.15420.

3. Kingsley, Yeon,, Promit Ghosal, and Mihai Anitescu. "BOLT: Block-Orthonormal Lanczos for Trace estimation of matrix functions." arXiv (Cornell University) (2025). doi:10.48550/arxiv.2505.12289.

4. Yeon, Kingsley, and Mihai Anitescu. "Beyond Low Rank: Fast Low-Rank + Diagonal Decomposition with a Spectral Approach." arXiv (Cornell University) (2025). doi:10.48550/arxiv.2512.17120.

5. Ramadan, Mohammad S., Evan Toler, and Mihai Anitescu. "Floodgates up to contain the DeePC and limit extrapolation." arXiv (Cornell University) (2025). doi:10.48550/arxiv.2501.17318.

6. Chen, Qi, and Mihai Anitescu. "Fourier-Enhanced Recurrent Neural Networks for Electrical Load Time Series Downscaling." arXiv (2025). http://arxiv.org/abs/2512.07876v1.

7. Montoison, Alexis, François Pacaud, Sungho Shin, and Mihai Anitescu. "GPU Implementation of Second-Order Linear and Nonlinear Programming Solvers." arXiv (2025). http://arxiv.org/abs/2508.16094v2.

8. Niu, Zehao, Mihai Anitescu, and Jie Chen. "Neighborhood Sampling Does Not Learn the Same Graph Neural Network." arXiv (Cornell University) (2025). doi:10.48550/arxiv.2509.22868.

9. Kuang, Wei, Alexis Montoison, Vishwas Rao, François Pacaud, and Mihai Anitescu. "Recovering sparse DFT from missing signals via interior point method on GPU." arXiv (Cornell University) (2025). doi:10.48550/arxiv.2502.04217.

10. Ramadan, Mohammad S., Marfred Barrera, Mihai Anitescu, and Sylvia Herbert. "Simultaneous improvement of control and estimation for battery management systems." arXiv (2025). http://arxiv.org/abs/2507.21300v

11. Dinenis, Philip, Vishwas Rao, and Mihai Anitescu. "Weakly-Constrained 4D Var for Downscaling with Uncertainty using Data-Driven Surrogate Models." arXiv (Cornell University) (2025). http://arxiv.org/abs/2503.02665v

12. Ramadan, Mohammad S., and Mihai Anitescu. "Dampening parameter distributional shifts under robust control and gain scheduling." arXiv (Cornell University) (2024). doi:10.48550/arxiv.2411.16566.

13. Zhao, Hongli, Tyler E. Maltba, Daniel Adrian Maldonado, Emil M. Constantinescu, and Mihai Anitescu. "Data-Driven Estimation of Failure Probabilities in Correlated Structure-Preserving Stochastic Power System Models." arXiv (Cornell University) (2024). doi:10.48550/arxiv.2401.02555.

14. Shin, Sungho, Wallace Gian Yion Tan, and Mihai Anitescu. "Improved Approximation Bounds for Moore-Penrose Inverses of Banded Matrices with Applications to Continuous-Time Linear Quadratic Control." arXiv (2024). http://arxiv.org/abs/2411.04400v2.

15. Ni, Runxin, Wang, Haoxuan, Sen Na, Sungho Shin, and Mihai Anitescu. "Parallel Sequential Quadratic Programming with Overlapping Graph Decomposition and Exact Augmented Lagrangian." arXiv (Cornell University) (2024). doi:10.48550/arxiv.2402.17170.

16. Yarahmadi, Somayeh, Daniel Adrian Maldonado, Lamine Mili, Junbo Zhao, and Mihai Anitescu. "Voltage-Dependent Modeling of Electromechanicalwave Propagation for Power System Dynamicstability Analysis." 2024. doi:10.2139/ssrn.4755587.

17. Luan, Lele, Nesar Ramachandra, Sandipp Krishnan Ravi, Anindya Bhaduri, Piyush Pandita, Prasanna Balaprakash, Mihai Anitescu, Changjie Sun, and Liping Wang. "Application of probabilistic modeling and automated machine learning framework for high-dimensional stress field." arXiv (Cornell University) (2023). doi:10.48550/arxiv.2303.16869.

18. Attia, Ahmed, Daniel Adrian Maldonado, Emil M. Constantinescu, and Mihai Anitescu. "Centralized calibration of power system dynamic models using variational data assimilation." arXiv (Cornell University) (2023). doi:10.48550/arxiv.2311.07676.

19. Maldonado, Daniel Adrian, Emil M. Constantinescu, Junbo Zhao, and Mihai Anitescu. "Efficient Computation of Power System Maximum Transient Linear Growth." arXiv (Cornell University) (2023). doi:10.48550/arxiv.2302.10388.

20. Niu, Zehao, Mihai Anitescu, and Jie Chen. "Graph Neural Network-Inspired Kernels for Gaussian Processes in Semi-Supervised Learning." arXiv (Cornell University) (2023). doi:10.48550/arxiv.2302.05828.

21. Lam, Albert, Mihai Anitescu, and Anirudh Subramanyam. "Network Cascade Vulnerability using Constrained Bayesian Optimization." arXiv (Cornell University) (2023). doi:10.48550/arxiv.2304.14420.

22. Yarahmadi, Somayeh, Daniel Adrian Maldonado, Lamine Mili, Junbo Zhao, and Mihai Anitescu. "Voltage-Dependent Electromechanical Wave Propagation Modeling for Dynamic Stability Analysis in Power Systems." arXiv (Cornell University) (2023). doi:10.48550/arxiv.2311.12326.

23. Pacaud, François, Sungho Shin, Michel Schanen, Daniel Adrian Maldonado, and Mihai Anitescu. "Condensed interior-point methods: porting reduced-space approaches on GPU hardware." arXiv (Cornell University) (2022). doi:10.48550/arxiv.2203.11875.

24. Subramanyam, Anirudh, Youngdae Kim, Michel Schanen, François Pacaud, and Mihai Anitescu. "A Globally Convergent Distributed Jacobi Scheme for Block-Structured Nonconvex Constrained Optimization Problems." arXiv (Cornell University) (2021). doi:10.48550/arxiv.2112.09027.

25. Yu, Jing, and Mihai Anitescu. "Solving Optimal Experimental Design with Sequential Quadratic Programming and Chebyshev Interpolation." arXiv (Cornell University) (2019). doi:10.48550/arxiv.1912.06622.

26. Matthews, Charles H., Bradly C. Stadie, Jonathan Weare, Mihai Anitescu, and Christopher L. DeMarco. "Simulating the stochastic dynamics and cascade failure of power networks." arXiv (Cornell University) (2018). doi:10.48550/arxiv.1806.02420.

27. Haley, CL, CJ Geoga, A Siegel, and M Anitescu. "Spectral analysis for comparison of second-order flow structure in DNS simulations." 2017.

28. Graziani, C, TJ Loredo, and M Anitescu. "Adaptive design of computer experiments and simulation fidelity hierarchies." preparation (2012).

29. Lockwood, B, and M Anitescu. "Use of Universal Gradient Enhanced Kriging Model for Regression Model Improvement and Error Estimation." 2010.

30. Anitescu, Mihai, William Layton, and Faranak Pahlevani. "Unconditional Stability for Numerical Scheme Combining Implicit Timestepping for Local Effects and Explicit Timestepping for Nonlocal Effects." arXiv (Cornell University) (2003). doi:10.48550/arxiv.math/0311363.

SPECIAL ISSUES EDITED

1. Mihai Anitescu and Goran Lesaja, Editors. Optimization Methods and Software Special Issue: "Special issue of Optimization Methods & Software in honour of Professor Florian A. Potra 60th Birthday", 2012.

2. Mihai Anitescu and Robert Fourer, Editors. Mathematical Programming B Special Issue: "The 20-th International Symposium on Mathematical Programming", 2009.

SOFTWARE

1. Kim, Youngseok, Peter Carbonetto, Mihai Anitescu, and Matthew Stephens. "mixsqp: Sequential Quadratic Programming for Fast Maximum-Likelihood Estimation of Mixture Proportions." CRAN: Contributed Packages (2018). doi:10.32614/cran.package.mixsqp.

OTHER

1. Negruţ, Dan, Laurent O. Jay, Alessandro Tasora, Mihai Anitescu, Hammad Mazhar, Toby Heyn, and Arman Pazouki. "NSF GRANT # 0700191 NSF PROGRAM NAME: CMMI Simulation of Multibody Dynamics Leveraging New Numerical Methods." 2011.

2. Boyd, Stephen, Real-time Embedded, Éva Tardos, and Mihai Anitescu. "Today's Plenaries." 2009.